

Physicochemical and biochemical characterization of antimicrobial peptides

Valentina Quezada^{a,*}, Paula Guzmán-Satoque^{a,*}, María Camila Rincón-García^a, Luis H. Reyes^b, and Juan C. Cruz^a

^aDepartment of Biomedical Engineering, Universidad de los Andes, Bogotá, Colombia, ^bDepartment of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia

1 Introduction

Antimicrobial peptides (AMPs) have garnered considerable attention as a promising class of antibiotics due to their diverse mechanisms of action, which differ from those of conventional antibiotics (Aoki & Ueda, 2013). These peptides disrupt membrane structure, inhibit protein and DNA synthesis, and repress cellular processes such as protein folding, cell wall synthesis, and metabolic turnover (Brogden, 2005; Park et al., 2011). Their multifaceted mechanisms enable them to exhibit strong antimicrobial activity against a wide range of microorganisms, including Gram-positive and Gram-negative bacteria, fungi, and viruses, often in the nanomolar or micromolar range (Wakabayashi et al., 1996; Yamauchi et al., 1993). Additionally, AMPs are effective against drug-resistant pathogens (Mygind et al., 2005). Unlike traditional antibiotics, which primarily kill bacterial cells or inhibit growth, AMPs possess a dual mechanism (Brunetti et al., 2016). Alongside potent antibacterial activity involving rapid membrane disruption (Lee et al., 2016), they also demonstrate immunomodulatory properties by activating pro- and antiinflammatory responses, chemoattraction, cell differentiation, innate and adaptive immune system activation, wound healing, and apoptosis (Mansour et al., 2014).

A broad understanding of AMPs can be gained through a deep analysis of their physicochemical and biochemical properties. These properties can clarify the peptides' mechanism of action and can be associated with their effectiveness. The main goal of physicochemical analyses is to establish basic characteristics such as molecular weight, amino acid composition, and secondary and tertiary structures. Biochemical studies, on the other hand, aim to describe peptide characteristics in a more biological context and provide insights into behaviors that occur both in vivo and in vitro, including stability, bioavailability, and pharmacokinetics analysis. Such comprehensive characterization is essential because, in spite of their enormous potential, AMPs are limited in clinical applications by a number of factors, such as their vulnerability to enzymatic degradation, susceptibility to deactivation in high-salt environments, and vulnerability to degradation in bodily fluids (Wang et al., 2017).

This chapter first examines the physicochemical characterization of AMPs. Techniques employed for molecular weight determination, amino acid composition, peptide sequence, secondary and tertiary structure, hydrophobicity and hydrophilicity, charge and pH dependence, solubility, thermal stability, and aggregation and oligomerization analysis will be described. In this comprehensive section, a wide array of techniques will be explored, including those based on chromatography (e.g., ion exchange chromatography, gas chromatography, high-performance liquid chromatography), spectrometry (e.g., mass spectrometry, UV spectrophotometry, Raman spectroscopy), and crystallography (e.g., X-ray crystallography). Computational methods will also be considered.

After elucidating the physicochemical characterizations, we will describe the main biochemical characterizations of AMPs. The section will aboard peptide's in vivo pharmacokinetic studies to determine pharmacokinetic parameters such as clearance, elimination half-life, $T_{\max}$, and $C_{\max}$. Moreover, in vitro bioavailability and stability studies will be explored to provide valuable insights into the enzymatic, pH, temperature, and serum stability of AMPs, along with chemical modifications to improve their bioavailability. Peptide-lipid interaction studies will also be considered as techniques to explore their mechanism of action. Then some receptor affinity assays employed to elucidate AMP interactions with cellular targets

*These authors contributed equally to this work.

(e.g., radioligand binding assays, surface plasmon resonance, fluorescence polarization) will be aboard. Finally, research highlighting the dual role of AMPs in terms of the antiinflammatory and immunomodulatory properties of peptides will be reviewed. A general schematic of the chapter content is presented in [Fig. 12.1](#).

2 Physicochemical characterization

2.1 Molecular weight determination

The determination of a peptide's molecular weight stands as a pivotal analytical parameter, crucial for devising methodologies aimed at its isolation from peptide mixtures and elucidating the appropriate protocols for characterizing its physicochemical attributes. Predominantly, experimental techniques such as SDS-PAGE (sodium dodecyl sulfate polyacrylamide gel electrophoresis) ([Judd, 2009](#)), high-performance liquid chromatography (HPLC) ([Mant et al., 2007](#)), and mass spectrometry ([Zhang et al., 2010](#)) are employed for this purpose. Nonetheless, the realm of bioinformatics offers expedited predictive capabilities concerning molecular weight estimation.

2.1.1 Gel electrophoresis

Gel electrophoresis techniques can determine the relative molecular mass of peptides. The methodologies employed are akin to those utilized for proteins. Nonetheless, certain modifications are implemented owing to the smaller size of peptides.

SDS-PAGE (sodium dodecyl sulfate polyacrylamide gel electrophoresis)

Sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) is a widely employed technique for the analytical separation of peptide mixtures with high resolution ([Judd, 2009](#)). It is a method primarily for providing information on the molecular weight and relative abundance of peptides. SDS functions as an anionic detergent, imparting a negative charge to peptides in proportion to their molecular mass ([Nowakowski et al., 2014](#)). Additionally, SDS denatures peptides, substantially eliminating secondary and tertiary structures. The SDS-PAGE method was introduced by Laemmli in 1970 ([Laemmli, 1970](#)).

The system proposed by Laemmli involves the use of a stacking gel and a resolving gel, both prepared with Tris-HCl and distilled water. The stacking gel is characterized by a low concentration of acrylamide (large pores) and a low pH to concentrate peptides before size-based separation. The resolving gel has a higher pH than the stacking gel and a greater acrylamide concentration, facilitating the separation of peptides based on their size. Typically, gels between 10% and 15% are used to separate peptides ranging from 16 to 45 kDa. Additionally, 2-mercaptoethanol (2-ME) or dithiothreitol (DTT) are used to reduce disulfide bonds.

The separation of peptides with a mass below 3 kDa ([Rivera et al., 2018](#)) is challenging using the basic SDS-PAGE protocol since Laemmli's procedure facilitates the separation of peptides over 10 kDa. Furthermore, the presence of native random-coiled small peptides adds to the challenge of separating them based on molecular size using the conventional SDS-PAGE method ([Sarfo et al., 2003](#)). Therefore, modifications have been proposed, including the addition of Urea or Tricine, namely Urea-SDS/PAGE and Tricine-SDS/PAGE, along with high concentrations of polyacrylamide and cross-linking agents to enhance the resolution and separation of very small peptides. It has also been reported that the separation of small peptides depends on the presence of SDS, as it enables their detection in the gel ([Sarfo et al., 2003](#)).

Native PAGE (polyacrylamide gel electrophoresis)

Native PAGE (polyacrylamide gel electrophoresis) is a seldom-used technique for the separation of peptides ([Arndt et al., 2019](#)), particularly those with a molecular mass below 1 kDa. This is because the separation is based on the difference in size and charge of peptides ([Kügler et al., 1997](#); [Ly & Wasinger, 2008](#)). Therefore, the preference lies in the use of liquid-based electrophoretic techniques ([Wittig & Schagger, 2008](#)). In the case of proteins, the native PAGE technique is employed because it does not denature proteins, allowing for the determination of protein-protein interactions. Indeed, the utilization of native PAGE is favored in the analysis of proteins due to its ability to preserve the higher-order structure. In contrast to denaturing techniques such as SDS-PAGE, native PAGE allows for the separation of proteins without disrupting their native conformation ([Nowakowski et al., 2014](#)). Therefore, for size-based separation of peptides, the preference lies in the utilization of Urea-SDS/PAGE and Tricine-SDS/PAGE, as they yield superior resolution.

The emphasis has been placed on the use of native PAGE for studying interactions between peptides and nucleic acids, coupled with complementary techniques such as size exclusion chromatography (SEC) in conjunction with circular dichroism (CD) ([J. Wang et al., 2023](#)). Additionally, the native PAGE technique is also employed in conjunction with

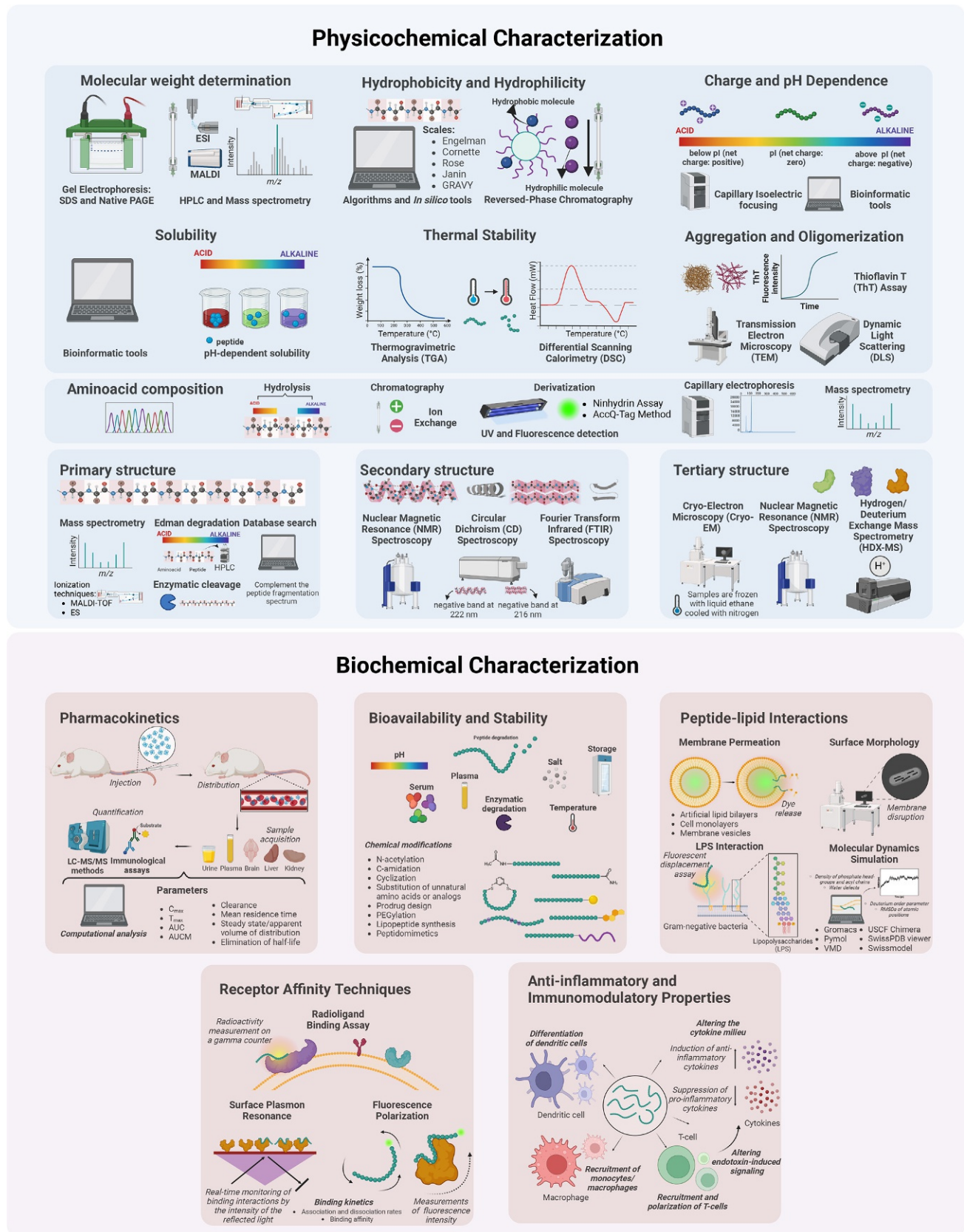


FIG. 12.1 Physicochemical and biochemical characterizations. General schematic of the physicochemical and biochemical characterization of antimicrobial peptides. The figure has two main sections. The first one describes the physicochemical techniques employed in the characterization of peptides. The second one describes the biochemical techniques employed in the characterization of antimicrobial peptides.

SDS-PAGE for peptide studies. In conclusion, if the goal is to separate a mixture of peptides and estimate their molecular mass using gel electrophoresis, the recommended approach is the use of SDS-PAGE.

2.1.2 High-performance liquid chromatography (HPLC)

High-performance liquid chromatography (HPLC) is a widely employed technique for the separation of peptides due to its efficiency and speed. Traditionally, HPLC forms such as size exclusion chromatography (SEC), ion-exchange chromatography (IEX), and reversed-phase HPLC (RP-HPLC) are commonly utilized for peptide separation (Mant et al., 2007). This technique is often coupled with others to determine the molecular weight of peptides.

Size exclusion chromatography (SEC)

Size exclusion chromatography (SEC) is a gel chromatography technique that allows for the separation of molecules, particularly peptides, based on their molecular size. This technique was first introduced in the late 1950s, following the discovery that proteins could be separated in cross-linked dextran porous gels. Various names are used to refer to this technique, including gel-filtration, steric exclusion, gel chromatography, gel filtration chromatography, gel exclusion chromatography, gel permeation chromatography, size separation chromatography, and molecular sieve chromatography (Janča, 2005).

In SEC, columns are packed with a gel filtration matrix with a defined pore size. Larger molecules will elute first, as they cannot penetrate as many pores, resulting in a shorter retention time compared to smaller molecules that can access more pores. Cross-linked gels made of dextran, agarose, polyacrylamide, and porous glass beads have been employed in SEC (Sun et al., 2011).

The determination of molecular weight distributions can be achieved using calibration standards to establish the relationship between elution volume and molecular weight. However, to avoid the use of standards, a simultaneous measurement of the intensity and concentration of the eluted fraction's light scattering can be conducted to estimate the average molecular weight (Grcev et al., 2004). The relationship between the light scattering intensity of a molecule and its molecular weight and size is explained by Rayleigh's theory, which posits that the molecular weight of a molecule is proportional to the ratio of the scattered light intensity to the incident light intensity (Stetefeld et al., 2016). Therefore, the molecular weight of peptides, previously separated based on size by SEC, can be estimated independently of elution volume (Oliva et al., 2001). Recently, the SEC technique has been employed for conducting proteolysis studies (Duijsens et al., 2023).

However, this methodology must be specifically adapted and designed for the separation of very small peptides. Additionally, the choice of elution type and flow rate must be carefully selected (Li et al., 2022).

2.1.3 Mass spectrometry (MS)

Mass spectrometry (MS) is a highly valuable technique in the analysis of peptides and proteins, owing to its capacity to yield a diverse array of information in a rapid, sensitive, and versatile manner. This tool facilitates the measurement of the molecular weight of peptides with an accuracy of 0.01%, in contrast to conventional methods such as SDS-PAGE, which exhibit an accuracy range of 5%–10%. In general, mass spectrometry allows for the determination of both the relative and absolute quantities of proteins, even in complex samples (Zhang et al., 2010).

Mass spectrometers are primarily composed of an ion source, a mass analyzer, and a detector. The most common ionization methods are MALDI (matrix-assisted laser desorption/ionization) and electrospray. Ions are separated based on their mass-to-charge ratio (m/z) in the mass analyzer and subsequently detected. The resulting graph is a mass spectrum, where the abundance of each ion is plotted against its m/z value (Zhang et al., 2010; Shuken, 2023; Foreman et al., 2021; Phetsanthad et al., 2023).

Matrix-assisted laser desorption/ionization (MALDI) MS

The matrix-assisted laser desorption/ionization (MALDI) technique was introduced in the late 1980s by Karas and Hillenkamp (Cournut et al., 2023). Its underlying principle involves subjecting samples to bombardment by ultraviolet (UV) photons. The matrices employed are specifically designed to exhibit strong absorption of UV light at a particular laser wavelength.

Typically, the sample should be at a low concentration and mixed with a matrix solution at a relatively high concentration (Strupat, 2005). The accuracy of mass determination through MALDI-MS is generally within the range of 0.5–2 ppm, a level of accuracy comparable to that of ESI-MS (Strupat, 2005).

The matrices commonly employed in matrix-assisted laser desorption/ionization (MALDI) include α -cyano-4-hydroxycinnamic acid (α -cyano or CHCA), 3,5-dimethoxy-4-hydroxycinnamic acid (sinapinic acid or SA), and 2,5-dihydroxybenzoic acid (DHB). CHCA is the most widely used matrix for peptides. The variation among these matrices lies in the amount of energy they impart to peptides and proteins, ultimately leading to fragmentation (Zhang et al., 2010). The quantification of both relative and absolute abundance is achieved through matrix-assisted laser desorption/ionization (MALDI) and tandem mass spectrometry (MS/MS) techniques.

MALDI-MS offers notable advantages, including speed, accuracy, high resolution, robustness, and sensitivity. Additionally, it exhibits greater tolerance toward impurities compared to ESI-MS at flow rates of $\mu\text{L}/\text{min}$ (Strupat, 2005). However, its limited utility in the analysis of very small peptides is attributed to the interference posed by low-molecular-weight peptides with the MALDI matrices employed for ionization (Heres et al., 2021). The preference lies in the utilization of techniques such as MALDI-TOF, whose sensitivity and resolution are significantly higher when dealing with small peptides and proteins (Zhang et al., 2010).

Electrospray/ionization (ESI) MS

Ionization is achieved by placing the analyte (peptide or protein) dissolved in a metal capillary (Zhang et al., 2010). Generally, samples, peptides, or proteins are dissolved in a water/methanol (1:1) or water/acetonitrile (1:1) solution (Strupat, 2005). The sample concentration should be 1 pmol/L (Strupat, 2005). An electric field is applied to the samples to generate a mist of highly charged droplets containing the analyte molecules. During this process, the volatile solvent evaporates, allowing only the ions to reach the mass analyzer (Strupat, 2005).

The ESI mass spectrum comprises a series of signals, each differing in charge. The distribution of charge states is achieved through protonation or deprotonation. The number of protons observed on peptides depends on the peptide's mass and the number of residues. Typically, peptides are observed with 2 to 5 protons, and the m/z ratios are typically between $m/z = 300$ and $m/z = 1500$ (Strupat, 2005).

2.1.4 Bioinformatic tools

Indeed, there are bioinformatics tools that enable the prediction of the molecular weight of peptides, such as ExPASy ProtParam (<https://web.expasy.org/protparam/>). This program calculates the molecular weight based on the sum of the average isotopic masses of amino acids in the protein/peptide and the average isotopic mass of a water molecule.

2.2 Amino acid composition

The amino acid analysis allows us to determine the amino acid composition of both peptides and proteins, and it involves first the hydrolysis of peptide bonds to obtain individual amino acids, their separation, usually by chromatography, and their determination and quantification (Rutherford & Gilani, 2009). Moreover, throughout this process, it is crucial to routinely implement a calibration standard that consists of a known amino acid's solution with which a calibration curve can be performed (Rutherford & Gilani, 2009).

2.2.1 Hydrolysis

One of the first steps in amino acid analysis is the hydrolysis of the samples, as its purpose is to break the peptide bonds that hold amino acids together within a peptide or protein (Rutherford & Gilani, 2009), which can be achieved by acidic, basic, or enzymatic treatment (Fountoulakis & Lahm, 1998; Lamp et al., 2018). Acidic hydrolysis usually consists of incubating the samples for 20 to 24 h in a high-molar hydrochloric solution (HCl) at 110°C (Lamp et al., 2018; Rutherford & Gilani, 2009), or employing sulfonic acids such as methanesulfonic acid, which, in comparison with HCl, allows the determination of methionine sulfoxide and tryptophan as the last one is destroyed during HCl hydrolysis (Fountoulakis & Lahm, 1998). That's how various amino acids, including serine, threonine, cysteine, tryptophan, and methionine, cannot be determined quantitatively during acid hydrolysis as a fraction is always lost in the process (Lamp et al., 2018; Rutherford & Gilani, 2009). As an alternative, alkaline hydrolysis is mostly employed when tryptophan determination is important, as this amino acid is stable in basic solutions, and enzymatic hydrolysis is commonly used for the determination of asparagine, glutamine, and other sensitive residues that are otherwise destroyed during chemical hydrolysis (Fountoulakis & Lahm, 1998). That's why other methods such as antioxidant and scavenging agent addition, vapor-phase hydrolysis, and microwave hydrolysis are employed for amino acid recovery (Rutherford & Gilani, 2009).

2.2.2 High-performance liquid chromatography (HPLC)

On the other hand, HPLC refers to a form of liquid chromatography (LC) in which components of a sample within the liquid mobile phase are resolved according to their properties, such as shape, size, charge, affinity with the stationary phase, and hydrophobicity, all packed within a column (Bloch et al., 2010). This technique differs from conventional LC in the fact that in the case of amino acid analysis, amino acids are separated at a high pressure, which results in mobile phase velocities that are up to 10–60 times greater than conventional chromatographic techniques (Bloch et al., 2010). Currently, it is still one of the most common procedures used for both quantitative and qualitative amino acids analysis due to its short analysis times, high resolution, versatility, and efficiency (González de Llano & Polo Sánchez, 2003; Xu et al., 2020), as well as its wide range of applications, which include the study and characterization of biological samples' peptides and their residue content, for example, microalgae (Araya et al., 2021), bacteriocins from *Lactococcus lactis* (Tasakis & Touraki, 2018), and even animal-derived foods (Dai et al., 2014) among many other applications. This method can adapt itself for the analysis of a great variety of different peptides and proteins, as the stationary phase, pH, aqueous buffer's ionic phase, type of organic modifier, gradients, and temperatures can all be modified accordingly to the sample (González de Llano & Polo Sánchez, 2003). Also, it allows the possibility of using microcolumns that have helped to detect quantities on the order of picomoles, which is an important step toward HPLC-mass spectrometry coupling (González de Llano & Polo Sánchez, 2003).

2.2.3 Ion exchange chromatography

Ion exchange chromatography (IEC) is an LC method that has been widely used for protein, peptide, antibody, and amino acid separation due to its high capacity, moderate cost, and easy control of the separation process (Cummins et al., 2017; González de Llano & Polo Sánchez, 2003; Kent, 1999; Velickovic et al., 2012; Wan, 2021). It was first developed by Moore and Stein in 1951, using a pH gradient to separate amino acids (Rutherford & Gilani, 2009). IEC's technique is based on Coulomb interactions between amino acids, peptides, and proteins to oppositely charged moieties linked covalently to the stationary phase or matrix of the chromatography column (Velickovic et al., 2012; Wan, 2021). This is possible because amino acids, peptides, and proteins are amphoteric compounds, which means that their net charge changes based on the pH of the solution. At a pH level above its isoelectric point (pI), the analyte becomes negatively charged (anion exchange), and at a level below its pI, it becomes positively charged (cation exchange) (Cummins et al., 2017; Velickovic et al., 2012). Therefore, IEC's experimental design allows the modification of both the strength of the interactions between the matrix and the analyte and the ionic properties of ionizable functional groups of both phases by changing the ionic strength of the solution and by pH titration of either the matrix or the analyte, respectively (Velickovic et al., 2012). Moreover, its configuration allows the selection of the ion-exchange matrix, elution format and buffer, adsorption, and elution conditions (Cummins et al., 2017). For example, it has been used for the study and characterization of biopharmaceutical proteins and peptides (Fekete et al., 2015) such as peptide samples with antibacterial properties (Dionysius & Milne, 1997; Raharjo et al., 2021; Recio & Visser, 1999).

2.2.4 Derivatization

Derivatization is a chemical process by which it is possible to increase amino acids sensitivity to both UV and fluorescence detection since most amino acids lack UV or fluorescence absorption functional groups (Xu et al., 2020). It is commonly used because it improves the chromatographic characteristics of an analyte by increasing the analyte's volatility, decreasing its polarity, and as previously mentioned, increasing the sensitivity of the detector to the study's analytes (González de Llano & Polo Sánchez, 2003). The most commonly used derivatization reagents are: ninhydrin, o-phthalaldehyde (OPA), phenylisothiocyanate (PITC), 6-aminoquinolyl-*N*-hydroxysuccinimidyl carbamate (AQC), alkyl chloroformate, fluorescein isothiocyanate (FITC), *N*-methyl-*N*-(trimethylsilyl)trifluoroacetamide (MSTFA) and fluorenylmethyl chloroformate (Fmoc-Cl) (Kaspar et al., 2009; Xu et al., 2020). It can be performed both pre- and postcolumn, which means if the derivatization of the sample is performed before or after column chromatography, the prior one is the most used for LC (Herráez-Hernández & Campíns-Falcó, 2000). According to Bardelmeijer et al. (1997), in postcolumn derivation, side reactions are not important and a complete derivation is not always necessary, but it has been shown to be a method with limited options of reaction conditions and possible reagent interference during detection (Bardelmeijer et al., 1997; Rutherford & Gilani, 2009). On the other hand, precolumn derivation includes a broader range of options for reagents and reaction conditions, as well as no limit on reaction kinetics. However, one of its main disadvantages is its possible interference in later chromatographies due to side products or unreacted reagents (Bardelmeijer et al., 1997; Rutherford & Gilani, 2009).

Ninhydrin assay

Ninhydrin is a derivatization reagent that can be used postcolumn for amino acid detection (Macchi et al., 2020). Amino acids form a purple complex with Ninhydrin at pH 8 and under heating (100°C–120°C) or derivatization conditions as it reacts with primary amines that include the α -amino group of all amino acids and the ϵ -amino group of lysine to produce Ruhmann's purple (diketohydrindylidenediketohydrindamine) (Rutherford & Gilani, 2009; Zhang & Ruan, 2015). Therefore, color change and intensity can be measured with a spectrophotometer at 570 nm as a measurement of the total content of amino acids (Rutherford & Gilani, 2009; Zhang & Ruan, 2015). This assay is widely used to analyze and characterize proteins, peptides, amino acids, and other ninhydrin-positive compounds in clinical, biomedical (Dufresne et al., 2017), food (Nwachukwu & Aluko, 2019), microbiological (Soudy et al., 2012), and plant studies (Friedman, 2004; Rogozhin et al., 2020).

AccQ-tag method

AccQ-Tag method is a precolumn derivatization of residues with 6-aminoquinolyl-*N*-hydroxysuccinimidyl carbamate (AQC) (Kaspar et al., 2009). This method's principle is based on the conversion of peptides' and proteins' primary and secondary residues into fluorescent and stable derivatives that are susceptible to fluorescence, UV-absorbance, electrochemical, and mass spectrometric detection (Kaspar et al., 2009; Ng, Chan, et al., 2023). Currently, this process can be performed by using commercial kits (e.g., AccQ-Tag Ultra Derivatization) that contain borate buffer, reagent powder (which contains AQC), and reagent diluent (May et al., 2021). Moreover, the use of the AccQ-Tag derivatization kit, along with liquid chromatography and subsequent tandem mass spectrometry (LC-MS/MS), is used in the analysis of free standard amino acids in complex samples because it is selective, sensitive, fast, and reproducible (Ng, Chan, et al., 2023). Also, with the AccQ-Tag Ultra UPLC method, it has been possible to obtain results 3–5 times faster than with conventional HPLC as it employs columns packed with 1.7- μ m particles (Kaspar et al., 2009). AccQ-Tag has been used in various types of studies, including the analysis of biological samples such as plasma (Ng, Chan, et al., 2023) and disease-related samples (Armenta et al., 2010; Guba et al., 2022).

2.2.5 Gas chromatography (GC)

GC is a high resolving power chromatography technique that involves the separation of volatile components of a mixture as they pass within a gaseous mobile phase through a stationary phase or column (Rédei, 2016) and requires prior derivatization (Kaspar et al., 2009). The most common derivatization process employed is silylation, which consists on the replacement of an active hydrogen by an alkylsilyl group using *N,O*-bis-(trimethylsilyl)trifluoroacetamide (BSTFA) or *N*-methyl-*N*-(trimethylsilyl)trifluoroacetamide (MSTFA) (Kaspar et al., 2009). However, throughout this process, not all derivatives obtained are stable, and both the reagents and derivatives are sensitive to moisture (Kaspar et al., 2009). Some alternatives to silylation include both alkylation and esterification to replace an acidic proton by an alkyl group; acylation, in which a nonacidic proton is replaced by an acyl group, and condensation, which involves the replacement of two protons in the formation of a double bond or cyclic derivative (Kaspar et al., 2009; Wells, 1999; Zumwalt et al., 1970). On the other hand, GC offers a relatively simple means of identifying unknown amino acid compositions by combining them with mass spectrometry (Hušek & Macek, 1975).

2.2.6 Capillary electrophoresis (CE)

CE is a fast and simple electrophoresis technique with extremely high resolving power that allows the separation of peptides composed of different amino acids or differing in only one residue (González de Llano & Polo Sánchez, 2003) due to differences in their size and net charge, as these factors determine the migration behavior of peptides and their amino acids in an electric field (Stefanik et al., 2024). Moreover, it is a highly efficient, green, and cost-effective alternative technique to LC (Stefanik et al., 2024), that requires low sample and reagents consumption (Pérez-Míguez et al., 2019). This technique doesn't necessarily require previous derivatization of the amino acids, but a UV-active label may be introduced to increase the CE-UV analysis' sensitivity for which derivatization is employed even more so because a high number of amino acids do not have enough UV absorption to be detected (Kaspar et al., 2009; Pérez-Míguez et al., 2019). However, the main disadvantage of CE with UV photometry detection lies in its low sensitivity (LOD within the range 10^{-5} – 10^{-6} mol/L); therefore, fluorescence detection has been more widely implemented as it provides a higher sensitivity (Tůma, 2021). In fact, it is one of the most used techniques in the separation of peptides for food analysis (Dong, 1999; González de Llano & Polo Sánchez, 2003), in the analysis of biological samples (Pérez-Míguez et al., 2019), in the study of therapeutic peptides or clinical samples (Stefanik et al., 2024), and even in the determination of nonprotein amino acids (Pérez-Míguez

et al., 2019). Its application together with other analytical techniques such as HPLC and size exclusion chromatography (technique mainly used for peptide and protein separation from samples), as well as coupling with mass spectrometry, has been important for the characterization of complex peptide samples (González de Llano & Polo Sánchez, 2003). Currently, CE has different modes and variations, which include capillary zone electrophoresis (CZE), which is based on the behavior of the analyte in a conductive solution in an electric field (Pérez-Míguez et al., 2019).

2.2.7 UV spectrophotometry and mass spectrometry

UV Spectrophotometry is a detection method used to identify and determine the amino acid content of peptides. This method is a quantitative measurement of the transmission or absorbance properties of a material (e.g., amino acids) as a function of wavelength (Myers, 2019) within the UV spectrum. In particular, tryptophan, tyrosine, and phenylalanine have a suitable sensitivity to UV spectrophotometry in their native forms due to their aromatic moiety (Ferré et al., 2019). However, the remaining amino acids do not possess moieties with chromophore properties; therefore, a derivatization step is needed to make them detectable within the UV spectrum (Ferré et al., 2019). In addition, UV spectrometry is unable to differentiate amino acids only with their detector signal, which is why chromatographic separation is needed to obtain correct identification and quantification (Violi et al., 2020).

Alternatively, mass spectrometry is another detection method employed in amino acid analysis for peptide quantification (Qasrawi et al., 2023). It measures molecules' mass-to-charge (m/z) with high sensitivity (Mann et al., 2001) that can even reach the fmol range (Mirgorodskaya et al., 2019), and for which derivatization is no longer needed since mass is an intrinsic and mostly unique property of all amino acids (Ferré et al., 2019). Peptides produce fragments that provide information on their amino acid composition, especially from the product ions of side-chain losses and from the immonium ions ($^+NH_2=CH-R$), which are formed from amino acids and have a specific mass and stability depending on the side-chain structure (Medzihradszky, 2005). In other words, MS is a major tool for protein, peptide, and amino acid identification as it provides both m/z values and their intensities, which can then be compared to theoretical values of peptide and protein composition after fragmentation (Wysocki et al., 2005). In this sense, it should be noted that mass spectrometers might present difficulties distinguishing between isoleucine and leucine or between glutamyl and lysyl residues as they have the same nominal mass, cases for which the different retention time and small mass difference allow them to differentiate these amino acids, respectively (Mann et al., 2001). Therefore, MS has become the most widely used detection method for the detection of amino acids at very low concentrations in complex samples, especially when in combination with LC and CE (Ferré et al., 2019), due to its high selectivity and sensitivity for detecting both underivatized and derivatized amino acids (Qasrawi et al., 2023). And, in contrast with UV spectrometry, MS does not require previous chromatographic separation, as the overlapping of amino acids with different masses does not affect amino acid determination and qualification (Violi et al., 2020).

2.3 Peptide sequence analysis

Peptide sequence analysis refers to the process of identifying the amino acids that constitute the primary structure of a peptide. This study serves as the basis for understanding the structure, function, properties, and interactions of peptides.

2.3.1 Mass spectrometry (MS)

In MS protein studies, peptide fragmentation analysis allows for the extraction of information about peptide structure and its sequence, hence, it can be used for de novo sequencing, protein and peptide identification, and covalent posttranslational modifications (Medzihradszky, 2005). Also, MS offers two types of approaches for protein and peptide sequencing: top-down sequencing and bottom-up sequencing (Wysocki et al., 2005). On one hand, bottom-up sequencing detects the maximal number of overlapping peptides to create a complete peptide map (Forbes et al., 2004). This MS sequencing method requires the analysis of the MS/MS spectra of the resulting peptide fragments to undergo peptide identification and relies on the use of algorithms for amino acid sequence alignments and assignments (Wysocki et al., 2005). On the other hand, top-down sequencing involves direct peptide sequencing in the gas phase, in which the protein or peptide sample is not subjected to enzymatic digestion but instead is transferred into the gaseous phase intact (Wysocki et al., 2005). Through this method, the intact peptide's molecular weight is first determined, then the peptide is subsequently fragmented and is again measured, so that, in combination with databases, the peptide's map and sequence are determined (Forbes et al., 2004; Wysocki et al., 2005). This approach provides almost 100% sequence coverage of the peptide and improved detection of posttranslational modifications such as acetylation, phosphorylation, and others (Forbes et al., 2004; Medzihradszky, 2005; Wysocki et al., 2005).

This technique requires charged gaseous molecules for its analysis. That's why ionization techniques have been incorporated into this detection technique, which include matrix-assisted laser desorption ionization (MALDI-TOF) and electrospray (ES), which were both developed by the 1980s (Gogichaeva & Alterman, 2019; Mann et al., 2001). It is worth adding that MS, in all its various methods (MALDI-TOF, ESI-MS and MS/MS), has been widely used in peptide de novo sequencing for discovering AMPs from biological samples such as frog skin secretions (Liu et al., 2012; Samgina et al., 2023), venoms (Miyashita et al., 2017), ribosome samples (Moyer et al., 2021), and microorganisms' cultures (Ren et al., 2023).

MALDI-TOF (matrix-assisted laser desorption/ionization time-of-flight)

MALDI-TOF is one of the soft ionization methods that volatilizes biomolecules that arose from MS and is used in the analysis of peptide sequences. This method does not require chromatographic separation or amino acid derivatization (Gogichaeva & Alterman, 2019). In this technique, a large excess of protonated molecules of the organic matrix (usually with an aromatic ring structure that absorbs the wavelength of the laser) is coprecipitated or co-crystallized with the analyte to obtain the gaseous ionized sample of analysis (Mann et al., 2001; Wysocki et al., 2005). In addition, the term TOF indicates that the analyzers accelerate the ions by using a short voltage gradient and measure the time the ions take to pass through the flight tube so that the flight time is proportional to the square root of the m/z (Wysocki et al., 2005), therefore it is a high resolving power mass analyzer (Kaspar et al., 2009). Moreover, this technique does generate protein fragmentation, which is why it is mostly used in the analysis of peptides (Mann et al., 2001).

In addition, it is important to note that some peptide ions decay through the acceleration process inside the flight tube, which is used for obtaining structural information in a technique called postsource decay, but this information does not serve as a peptide sequencing method (Mann et al., 2001) unless it is high-quality, which can provide an illustration of de novo sequencing (Medzihradszky, 2005).

ESI-MS (electrospray ionization mass spectrometry)

ESI-MS is the other ionization method that arose from MS, is widely used for the analysis of peptide sequences, and is considered one of the most powerful and widely used detection methods (Ferré et al., 2019). Its mechanism of ionization is based on electrostatically dispersing the liquid analyte by pumping it at a low $\mu\text{L}/\text{min}$ flow rate and high voltage through a hypodermic needle (Mann et al., 2001). ESI is currently predominantly used in the analysis of low-molecular-weight molecules such as amino acids, hormones, and pharmaceuticals when in combination with HPLC or LC (Gogichaeva & Alterman, 2019; Yates et al., 1994), but both MALDI-TOF and ESI have demonstrated good comparability in quantitative analysis of low-molecular-weight compounds (Medzihradszky et al., 2000). However, ESI presents some disadvantages in terms of a possible competition between analytes for a charge during the ionization process in the spray droplets, which means that in a peptide digest analyzed without separating the peptides, only a reduced number of the most easily ionized peptides or amino acids will be detected (Wysocki et al., 2005).

Tandem mass spectrometry (MS/MS or MS2)

Peptide sequencing can be performed by interpreting the data results from the fragmentation spectra obtained (Wysocki et al., 2005) by performing two steps of MS in tandem, either employing the same MS particle separation principle twice or by combining two different separation principles (Mann et al., 2001). This method mostly replaced traditional nonfragmentation MS, while it added another level of improvement in analyte determination as it can select specific ions, induce their fragmentation, and measure the m/z of those fragmented ions, resulting in higher specificity and reproducibility (Neagu et al., 2022; Violi et al., 2020; Wysocki et al., 2005). In particular, MS/MS provides a sequence-specific fragmentation pattern of the analyte peptide, which can be used for sequence database mapping and matching (Taylor & Johnson, 2001). In this sense, the subsequent computer approach consists of calculating all possible amino acid sequences according to the peptide mass and then determining which of those sequences is the best match according to the MS/MS spectrum obtained, for which the software search engine Mascot (Taylor & Johnson, 2001) or PEAKS might be used (Ma et al., 2003). Currently, there are a wide variety of different MS/MS configurations in terms of speed, ionization method, resolution, m/z range, and sensitivity, which allow the sequencing of all types of peptides (Wysocki et al., 2005). In fact, peptide sequencing is mostly performed on electro-sprayed ions, as they generally have a charge that corresponds to the number of positively charged amino acids and the charge from the N terminus (Mann et al., 2001). Moreover, Tandem Mass Spectrometry is highly used for de novo sequencing of peptides with therapeutic applications such as cyclic peptides (Ngoka & Gross, 1999; Townsend et al., 2018) and AMPs (Liu et al., 2012; Samgina et al., 2023).

2.3.2 Edman degradation

Edman degradation was first proposed by Pehr Edman in 1950 and consists of a series of chemical reactions used to remove and determine the amino acid on the N terminus of a peptide (amino acid with a free α -amino group), so that subsequently the next amino acid on the sequence is analyzed through the same means until the complete sequence is identified (Smith, 2001). In this sense, some peptides and proteins have the amino group blocked by posttranscriptional modifications by the addition of acetyl, formyl, and pyroglutamyl groups (Smith, 2001). First, phenyl isothiocyanate (PITC) reacts with the free α -amino group of the residue at the N-terminal end of the polypeptide chain, and a phenylthiocarbamyl derivative of the terminal residue is formed under basic conditions (Smith, 2001). Then, in the presence of a strong acid, cleavage of the peptide bond holding the residue at the N terminus is done; it is liberated as the anilinothiazolinone (ATZ) form of the residue, and the shortened peptide can be separated in the organic phase to continue the same process (Smith, 2001). Finally, through the use of an aqueous acid, a more stable phenylthiohydantoin (PTH) form of the residue is obtained, which is then identified through chromatography. Later on, this process was automated through the use of an instrument termed sequenator (Smith, 2001). However, it is worth noting that new techniques such as mass spectrometry or combined methods such as LC-MS/MS have recently replaced Edman degradation for peptide sequencing due to their better resolution and sensitivity, whereas current applications are limited (Vecchi et al., 2019).

2.3.3 Enzymatic cleavage

Enzymatic cleavage, also known as enzymatic digestion, is another mechanism employed for peptide sequencing. To determine the de novo sequence of a peptide, enzymatic cleavage is first performed to break the peptide bonds at specific amino acids, which are then identified through amino acid analysis, most commonly through the use of LC-MS/MS (Zhang et al., 2023). Its main difference from chemical cleavage is the use of enzymes such as proteases to carry out selective peptide cleavage (Williams & Stone, 1997). For example, highly selective cleavage at Asn-Gly or Asp-Pro is being more widely used for partitioning peptides and proteins into tractable fragments (Ponstingl et al., 1982). It should be noted that selective proteinases are available for purchase and are named after the kind of bond they break and the amino acid. For instance, the endoproteinase Glu-C (V8 proteinase from *Staphylococcus aureus*) catalyzes the cleavage of the peptide bond at the C-terminal of glutamic acid (Matsumoto & Komori, 2000; Williams & Stone, 1997).

2.3.4 Ion mobility spectrometry (IMS)

IMS is a promising tool for applications in structural biology and proteomics, and it is also another technique used to separate and sequence peptides (Chang et al., 2021), proteins, protein complexes, and even explosives (Allen et al., 2016). In this technique, ions of different cross sections, m/z , and charge states are separated through collisions with a gas in a uniform electric field (Wysocki et al., 2005), in other words, it adds an extra dimension that allows the separation of molecules according to their size and shape in the gas phase (Ferré et al., 2019; Meier et al., 2021). In fact, its simplest form is an electric field that drives the ions through the cell in the presence of a gas, so that the electric field induces the separation of the amino acids, in this case by size and their interactions with the neutral gas particles (Ferré et al., 2019). More specifically, in IMS, ions migrate through an inert buffer gas under a weak electric field, where the analyte's collisions with the gas molecules retard the process by which the ions arrive at the detector, so that larger ions tend to have more collisions and subsequently arrive later than their smaller counterparts (Meier et al., 2021). In some cases, the analyte peptide or peptides are electro-sprayed through ESI before their IMS analysis to obtain their ionic form (Allen et al., 2016; Fernandez-Lima et al., 2011). Now, there is another technique of ion mobility spectrometry termed trapped ion mobility spectrometry (TIMS), which differs from the traditional IMS technique in that instead of driving the ions through a stationary gas, in TIMS ions are kept stationary in a moving gas column (Ridgeway et al., 2018). This modified IMS technique has shown to achieve high resolving power (R of about 300), and an efficiency of about 80% (Ridgeway et al., 2018).

These techniques can also be coupled with MS or MS/MS, which has shown to allow the effective characterization of isomeric biomolecules now based on their ion mobility-selected MS/MS spectrum (Chang et al., 2021; Fernandez-Lima et al., 2011), information of interest for areas such as pharmaceutical research, proteomics (including peptide studies), polymer science, environmental chemistry, medicine, and even archeology (Hurtado & O'Connor, 2012).

2.3.5 Database search

Peptide sequencing has become much simpler through database matching because, with de novo sequencing, the peptide fragmentation spectrum might not contain sufficient information to identify the complete amino acid sequence, and only a minimal fraction of all the possible amino acid sequences that might be obtained, though the spectrum does occur in nature

(Steen & Mann, 2004). Therefore, through database searching, it is possible to complement the peptide fragmentation spectrum and match this information to peptide and protein sequences that already occur in nature (Steen & Mann, 2004). Moreover, database search can be used for identifying peptides or even proteins based on sequence tag search and for de novo peptide sequencing, usually from data obtained via MS/MS and other techniques listed in this chapter.

Currently, there are plenty of software, algorithms, and search engines that can be employed for de novo sequencing of peptides based on databases of already known sequences and MS/MS spectra. For example, PEAKS is a software that can compute the most accurate peptide sequences whose fragment ions can best describe the results obtained from an MS/MS spectrum (Ma et al., 2003; Ng, Zhou, et al., 2023). Other examples of algorithms used for the same purpose of de novo peptide sequencing are Novo HMM, which is based on a Markov model (Fischer et al., 2005), and PepNovo, which is based on probabilistic network modeling (Frank & Pevzner, 2005; Ng, Zhou, et al., 2023). Furthermore, SEQUEST (Yates et al., 1995) and Mascot (Perkins et al., 1999; Taylor & Johnson, 2001) are the most used search engines for data processing after the respective amino acid analysis techniques utilized for peptide sequencing (Prakash et al., 2019). Bolt, a search engine for comprehensive MS/MS sequencing of peptides and proteins, is another example of how database search has been able to facilitate and speed up peptide sequencing up to a few minutes (Prakash et al., 2019).

2.4 Secondary structure analysis

The analysis of the secondary structure of peptides is crucial for determining their stability and predicting their tertiary structure (McGill et al., 2019). Various spectroscopic techniques are available for determining secondary structure, each with its advantages. The choice of technique depends on the specific goals of the analysis. For instance, circular dichroism (CD) spectroscopy is employed when a rapid determination is required. On the other hand, Raman or infrared spectroscopy is useful when dealing with small sample volumes (Pelton & McLean, 2000).

2.4.1 Nuclear magnetic resonance (NMR) spectroscopy

Nuclear magnetic resonance (NMR) spectroscopy provides atomic-resolution structural information for peptides and other macromolecules. It is a highly utilized technique for the analysis of the secondary structure of peptides and proteins. Depending on the presence of helical or beta-sheet conformations, shifts in the magnetic field occur in an upward or downward direction, respectively. One of the primary advantages of NMR is its ability to provide information about the location of a secondary structure within the peptide or protein sequence. This is achieved by assigning a chemical shift index (CSI) to each of the peptide or protein residues. Peptides can be analyzed without size limitations using this technique, as the high-resolution analysis limit extends to 30 kDa (Pelton & McLean, 2000).

2.4.2 Circular dichroism (CD) spectroscopy

Circular dichroism (CD) spectroscopy is a widely used technique for analyzing the secondary structure of peptides and proteins in solution. Molecules exhibit circular absorption of plane-polarized light in both right- and left-handed directions, resulting in elliptical polarization. Dichroism is measured as a function of wavelength, and spectral bands are associated with the structural features of peptides. Micro and nanoparticles are often employed as substrates to enhance the effective peptide concentration and ensure sufficient light absorption (North & Taitt, 2016). To perform the measurement, peptides or proteins are used in solution at a concentration ranging from 0.01 to 0.2 mg/mL (Pelton & McLean, 2000).

α -helices are observed at 222 nm with a negative band attributed to hydrogen bonding and a positive band between 190 and 195 nm. β -sheets exhibit a negative band around 216 nm, a positive band between 195 and 200 nm, and a negative band near 175 nm. Additionally, it is known that the average ellipticity can be used to estimate the content of α -helices (Pelton & McLean, 2000).

2.4.3 Fourier transform infrared (FTIR) spectroscopy

The principle of Fourier-transform spectroscopy is based on the absorption of energy resulting from the vibration of chemical bonds, involving both stretching and bending motions (Pelton & McLean, 2000). It is a widely used technique for the structural characterization of peptides and proteins. However, the analysis of the FTIR spectrum of a peptide is complex and challenging to interpret due to the inherent flexibility of peptides and the potential for additives, which can introduce noise in the amide I region (Adochitei & Drochioiu, 2011).

The amide I and II bands are the most representative vibrational bands. Amide I is typically observed between 1600 and 1690 cm^{-1} , representing the stretching of C=O bonds. Amide II is found between 1480 and 1575 cm^{-1} , corresponding to the

stretching of CN bonds and the bending of NH bonds. Additionally, amide III appears between 1229 and 1301 cm^{-1} , attributed to the stretching of CN bonds and the bending of NH bonds (Adochitei & Drochioiu, 2011).

Correlations can be established between the frequencies of infrared bands and the components of the secondary structure in peptides and proteins (Adochitei & Drochioiu, 2011). The position and intensity of these bands allow inferences regarding structural elements such as alpha helices and beta sheets. For example, in the amide I region, the wavenumbers 1624 cm^{-1} and 1635 cm^{-1} have been correlated with β -sheets, 1657 cm^{-1} with α -helices, 1648 cm^{-1} with random coils, and 1680 cm^{-1} with β -turns (Cobb et al., 2020).

2.4.4 Raman spectroscopy

Raman spectroscopy is a type of vibrational spectroscopy that involves inelastic scattering of light (Terentis & Ye, 2013). The Raman spectrum arises from the difference between the frequencies of incident and scattered radiation, and the utilized radiation is monochromatic. The radiation from a monochromatic light source ensures that almost all incident photons have the same energy as the scattered photons, corresponding to elastic scattering. Inelastic scattering occurs due to the energy shift of the scattered photons relative to the incident photons (Terentis & Ye, 2013). The exact wavenumbers of the bands depend on the hydrogen bonding patterns and the angles of the peptide bonds.

In Raman spectroscopy, the amide I and III bands are employed for the assignment and analysis of the secondary structure of peptides and proteins (Pelton & McLean, 2000). Assignment to specific secondary structure types is accomplished through reference spectra of proteins with a predominant structure, such as alpha helix or beta sheet, along with theoretical calculations and synthetic peptides (Pelton & McLean, 2000).

Fourier-transform Raman spectrometers can be employed to enhance signal-to-noise ratios, and longer-wavelength excitation sources can be utilized (Pelton & McLean, 2000). Amide I bands between 1650 and 1655 cm^{-1} are associated with α -helical conformations, while those between 1612 and 1640 cm^{-1} are linked to β -sheets (Pelton & McLean, 2000). Amide II bands between 1510 and 1530 cm^{-1} are correlated with antiparallel β -sheets, and between 1530 and 1550 cm^{-1} with parallel β -sheets (Pelton & McLean, 2000). However, a quantitative approach to determining the secondary structure involves deconvoluting the amide I band profile into a sum of component functions, typically Lorentzian or Gaussian-Lorentzian functions (Terentis & Ye, 2013). Additionally, confocal Raman microscopy can be employed to determine the structure of peptides in aqueous solutions (Terentis & Ye, 2013).

2.4.5 Computational methods

Computational methods are highly valuable for predicting protein structure. Various strategies are employed, with the majority based on homology or comparative modeling, as well as Ab Initio or de novo approaches (Majumder, 2020).

Homology-based techniques involve aligning the sequence of peptides or proteins with sequences of known peptides or proteins. The predicted structure is then based on the degree of similarity. Models are more reliable when the similarity percentage is higher (>50%) (Majumder, 2020). Some methods relying on this approach include RaptorX, Biskit, FoldX, Swiss-Model, PROTEUS, SABLE, Jpred, and HHpred (Majumder, 2020).

However, there are cases where homologous peptides or proteins cannot be identified, necessitating Ab Initio modeling based on the amino acid sequence of the peptide or protein. In this case, the model is generated based on energy considerations, specifically the most stable structure (Majumder, 2020). There are two types of energy functions: physics-based, considering physical and chemical properties, and knowledge-based, resolving statistically. Another consideration is the conformational search, which can primarily rely on two computational methods: Monte Carlo and Molecular Dynamics. Some methods using this approach include Falcon, Quark, Robetta, Abalone, YASSPP, and Bhageerath (Majumder, 2020; Oldfield et al., 2019). However, it's worth noting that these methods may be time-consuming, and there can be challenges with the orientation of amino acid side chains (Majumder, 2020).

Some of the most widely used predictors of protein structure are PSIPRED and PHD (Oldfield et al., 2019).

2.5 Tertiary structure analysis

The tertiary structure of a peptide corresponds to the three-dimensional conformation it can adopt based on the interaction between its secondary structural elements. Various types of bonds facilitate the formation of a stable tertiary structure, including hydrogen bonds, ionic bonds, and disulfide bridges. However, not all peptides exhibit a well-defined tertiary structure, as this depends on the peptide's length and other factors influencing folding and stability.

2.5.1 X-ray crystallography

X-ray crystallography is one of the most accurate and reliable methods for obtaining structural information about peptides and proteins. This method has been utilized for over 60 years (Spiliopoulou et al., 2020). The first successful crystallization was that of myoglobin in 1958 by Kendrew et al. (Reese et al., 2020). The principle of X-ray crystallography involves assessing the scattering of high-power X-rays on a crystalline structure, allowing the determination of atomic coordinates to deduce the 3D molecular structure (Reese et al., 2020). However, the major limitation of this technique is that not all proteins or peptides can be crystallized using traditionally established methods (Reese et al., 2020). For instance, peptides and proteins with hydrophobic character tend to aggregate and are challenging to crystallize. Additionally, peptides typically have fewer than 50 amino acids, which also makes them challenging to crystallize (Schroeder & Rosengren, 2020).

2.5.2 Nuclear magnetic resonance (NMR) spectroscopy

Nuclear magnetic resonance (NMR) spectroscopy emerges as a powerful technique facilitating the investigation of proteins and peptides, eliminating the necessity for crystallization. This method is particularly advantageous for conducting structural analyses on flexible proteins and peptides characterized by unstructured domains and small sizes (Hu, 2014). In general, NMR spectrometers with strong magnetic fields offer enhanced resolution. However, the utility of employing spectrometers with strong magnetic fields varies depending on the sample conditions (Schroeder & Rosengren, 2020). A fundamental prerequisite for conducting a detailed analysis of the tertiary structure through NMR is a spectrometer of >500 MHz (Schroeder & Rosengren, 2020).

One-dimensional nuclear magnetic resonance (1D NMR) is commonly employed for determining the structure of small molecules, such as small peptides. However, it is not as suitable for large peptides or proteins due to the substantial number of atoms, making the interpretation of the spectrum challenging (Schroeder & Rosengren, 2020). When conducting this type of analysis, it is crucial to assess whether the peptide signals are weak, as optimization may be necessary in such cases. In typical 2D NMR analyses, the peptide signals obtained from 1D NMR analysis are employed, along with experiments such as TOCSY, NOESY, and ROESY. Among these, ROESY proves to be the most beneficial for small peptides, especially when using a mixing time between 200 and 350 ms (Schroeder & Rosengren, 2020).

The determination of tertiary structure primarily involves the utilization of peptide sequence data and bond lengths and angles. This process results in an ensemble of randomly folded structures. By employing software such as CYANA or CNS, distance constraints between protons and torsion angles can be considered, allowing for the calculation of the structure based on NMR-derived data. The quality of the NMR structure can be validated by using tools such as PROCHECK and assessing factors such as NOE violations and energy favorability (Hu, 2014).

2.5.3 Cryo-electron microscopy (Cryo-EM)

Cryo-electron microscopy (Cryo-EM) is a technique that was implemented in 1975 to generate one of the first 3D models of the bacteriorhodopsin protein, which could not be crystallized (Cressey & Callaway, 2017). Cryo-electron microscopy allows for the deduction of the tertiary structure of biomolecules, including peptides and proteins, through the bombardment of electron beams. This technique can be employed for the analysis of proteins/peptides in solution, as it does not require sample dehydration (Carroni & Saibil, 2016). To analyze the 3D structure using Cryo-EM, samples are frozen with liquid ethane cooled with nitrogen and then transferred to the electron microscope. The entire environment is maintained at low temperatures using liquid nitrogen or helium.

In 2017, Jacques Dubochet, Joachim Frank, and Richard Henderson were awarded the Nobel Prize in Chemistry for the development of this technique (Cressey & Callaway, 2017). This technology can achieve a resolution between 3.5 and 4 Å. However, a limitation exists regarding the determination of true atomic positions, as a resolution of 1.5 Å is required (Yip et al., 2020). To address this, Yip et al. (2020) designed hardware with new electro-optic elements that enable the determination of protein or peptide structures with atomic resolution using Cryo-EM, achieving a structure of apoferritin with a resolution of 1.25 Å (Yip et al., 2020). Recently, this technique was used to analyze the structure of a peptide fibril that enhances retroviral infectivity (Heerde et al., 2023).

2.5.4 Hydrogen/deuterium exchange mass spectrometry (HDX-MS)

Hydrogen/deuterium exchange (HDX) has been a widely used tool for the structural analysis of peptides and proteins since the mid-20th century, and it has been coupled with mass spectrometry (HDX-MS) since 1990 (Ozohanics & Ambrus, 2020). It involves monitoring the exchange of amide protons with deuterium, requiring meticulous control of pH and temperature for this exchange reaction (Benhaim et al., 2019). Subsequently, pH (2.54) and temperature (0°C) are lowered to

trap deuterium in position. The kinetics of the reaction enable the determination of structural dynamics (Benhaim et al., 2019). Mass spectrometry is employed to quantify deuterium incorporation (Benhaim et al., 2019). Some advantages of this technique include its high sensitivity, low sample requirements, and applicability to the analysis of glycosylated peptides or membrane proteins. However, measures need to be taken to prevent deuterium loss, such as maintaining a very low pH and low temperatures (Benhaim et al., 2019).

For the mass spectrometry (MS), the analyzers commonly used for HDX include Orbitrap and quadrupole time-of-flight (QTOF) with electrospray ionization as the ion source. Desorption/ionization laser-assisted by matrix (MALDI) can also be used in conjunction QTOF analyzer (Ozohanics & Ambrus, 2020).

2.5.5 Small-angle X-ray scattering (SAXS)

Small-angle X-ray scattering (SAXS) is a technique that was initially employed in the 1930s for the analysis of metallic alloys (Edwards-Gayle & Wychowaniec, 2023). However, it has also been applied to the study of macromolecules in solution, including investigations into oligomeric states, temperature profiles, chemical equilibria, etc. (Edwards-Gayle & Wychowaniec, 2023). It is also utilized for the analysis of peptide and protein structures in solution, presenting an advantage as it provides information in a native state. However, one of the main limitations lies in the random orientation and movement of molecules in solution, preventing the establishment of spatial orientation for proteins and peptides. The approach proposed and implemented involves the use of SAXS data with Ab initio computational methods to generate 3D models (Prior et al., 2020).

Structural characterization with SAXS can be conducted on both ordered and disordered molecules, ranging from small kDa peptides to high molecular weight proteins. High-intensity synchrotron sources are typically employed (Kikhney & Svergun, 2015). The principle of the technique involves placing the peptide or protein solution in a quartz capillary and illuminating it with a collimated monochromatic X-ray beam, thereby collecting data on the intensity of X-ray scattering (Kikhney & Svergun, 2015).

2.6 Hydrophobicity and hydrophilicity

Hydrophobicity and amphipathicity are considered crucial parameters for peptides whose sole target is the cytoplasmic membrane.

2.6.1 Kyte-doolittle scale

The Kyte-Doolittle scale, proposed in 1982 by Kyte and Doolittle, serves as a straightforward approach to assess the hydrophobic or hydrophilic nature of a protein (Kyte & Doolittle, 1982). In the initial algorithm, implemented in the C language, hydrophobicity values were assigned to individual amino acid residues, and these values were subsequently summated, initiated from the amino-terminal end, aimed to calculate the average hydrophobicity. In general, the hydrophobicity of an amino acid is related to the transfer of free energy from a polar to an apolar environment (Koehler et al., 2009).

If the Kyte-Doolittle scale value is positive, then the peptide or protein region is considered hydrophobic. The characterization of peptide hydrophobicity is widely used and has been employed, for instance, in the analysis of pathogenic signal peptide variants in the human genome (Guarnizo et al., 2023).

Other scales that are used for similar purposes include the Engelman (Engelman et al., 1986), Eisenberg (Sweet & Eisenberg, 1983), Hopp-Woods (Hopp & Woods, 1981), Cornette (Cornette et al., 1987), Rose (Rose et al., 1985), Janin (Janin, 1979), and GRAVY (Xiang et al., 2017) scales.

2.6.2 Reversed-phase chromatography (RPC)

Reverse phase chromatography (RPC) is a technique utilized for the separation of peptides based on their hydrophobic nature, where the interaction between the stationary phase and solutes is controlled by changes in the polarity of the mobile phase (Al Musaimi, Valenzo, et al., 2023; Taneja et al., 1984). In 1984, Ashok and collaborators reported the use of size exclusion chromatography (SEC) and RPC for separating hydrophobic peptides (Taneja et al., 1984). In general, it is an accurate technique for small peptides (Al Musaimi, Valenzo, et al., 2023).

The nonpolar phase (resin) interacts with the hydrophobic residues of amino acids, enabling separation based on relative hydrophobicity (Al Musaimi, Valenzo, et al., 2023). This interaction is gradually destabilized with ethanol or methanol to influence the retention time (Al Musaimi, Valenzo, et al., 2023). However, various factors impact retention time, including the distribution of amino acids along the peptide sequence, overall charge, impurities, intrinsic structural characteristics of peptides, and peptide length (Al Musaimi, Valenzo, et al., 2023). Moreover, there are factors influencing the

hydrophobicity of amino acids that, in turn, impact the retention time. Some of these factors include the stationary phase, the composition of the mobile phase, sample size, and more (Al Musaimi, Mercado-Valenzo, et al., 2023).

Reverse phase chromatography (RPC) serves a dual purpose, functioning not only as a valuable tool for the separation of peptides but also as a method for determining the retention time. This retention time calculation demonstrates a linear dependency on the hydrophobicity of the peptides (Krokhin et al., 2004).

2.6.3 *In silico* tools

There are several *in silico* tools available to physiochemically characterize peptides. Hydrophobicity and hydrophilicity are commonly sought-after properties in peptides, as they provide insights into their interactions. Some of these tools include the Peptide Property Calculator, ExPASy, and ToxinPred (Rivero-Pino et al., 2023). Recently, the GUD-VE tool was developed, allowing for the comparison of physicochemical properties, such as the hydrophobicity, of different peptides/proteins (Chauhan et al., 2023).

2.6.4 *Experimental techniques: Fluorescence spectroscopy*

Fluorescence spectroscopy is a highly valuable tool for the structural and dynamic analysis of proteins/peptides. This technique is particularly attractive due to its straightforward data acquisition and analysis processes (Meyers, 2006). In the study of peptide/protein hydrophobicity, 8-anilino-1-naphthalenesulfonic acid (ANS) is employed to characterize the exposed hydrophobic groups of proteins/peptides (Guliyeva & Gasymov, 2020).

2.6.5 *Grand average of hydropathicity index (GRAVY)*

The grand average of hydropathicity (GRAVY) is a widely used measure to determine the hydrophobicity of a peptide or protein (Chang & Yang, 2013; Trejos et al., 2023; Wang, Zhong, et al., 2021). This index is calculated based on the hydropathy values of each amino acid comprising the peptide divided by the peptide length (Chang & Yang, 2013). Thus, if the value is positive, it is hydrophobic; conversely, if it is negative, it is hydrophilic (Chang & Yang, 2013).

2.7 Charge and pH dependence

The net charge of a peptide depends on the pH it is exposed to. At a pH higher than its isoelectric point (pI), the charge is negative, and at a pH lower than its pI, the charge is positive. Thus, the net charge is zero at its pI. The peptide's charge is determined by the pK values of ionizable groups (Pallbo et al., 2019; Shaw et al., 2001). Ionizable amino acid residues exposed to a solvent result in variable peptide charges depending on the surrounding pH. The study of electrostatic interactions is crucial as they influence the structure and function of proteins (Lošdorfer Božič & Podgornik, 2017).

2.7.1 *Isoelectric point (pI) determination*

The isoelectric point (pI) of a peptide can be determined through an isoelectric focusing approach using gels or capillaries (Štěpánová & Kašička, 2018). The capillary isoelectric focusing method offers higher resolution, sensitivity, and faster results (Štěpánová & Kašička, 2018). In this method, peptides are separated in a capillary based on their pI using a pH gradient. The capillary contains carrier ampholytes, the sample (peptides), and pI markers, followed by the application of a high voltage (Štěpánová & Kašička, 2018). The mobilization can occur by pressure or gravity while maintaining the electric field or by electrokinetic migration (Štěpánová & Kašička, 2018). pI markers are typically low molecular weight peptides or proteins with known isoelectric points. One limitation is the possibility of peptide precipitation, but this can be addressed by using glycerol, taurine, sorbitol, or sucrose (Audain et al., 2016; Štěpánová & Kašička, 2018).

There are various bioinformatics tools available for calculating the isoelectric point. In 2023, pIChemiSt was published as a free tool enabling the calculation of isoelectric points, even for modified peptides (Frolov et al., 2023). The principle of this tool for pI calculation involves identifying ionizable groups and predicting their pKa values (Frolov et al., 2023).

2.8 Solubility

The solubility of a peptide is crucial in determining the type of solvent to be used. However, as will be discussed later, solubility can depend on factors such as pH. Solubility can be determined experimentally using various types of solvents or through computational prediction tools.

2.8.1 Solubility prediction software

There are several bioinformatics tools available for predicting the solubility of peptides based on amino acid sequence and physicochemical properties. One example is MahLooL, a tool that does not rely on a server and is based on the use of recurrent neural networks to predict various physicochemical properties of peptides, including solubility (Ansari & White, 2023). Another need in science is the prediction of solubility for peptides containing modified amino acids (mAA), as experimental methods for measuring the physicochemical properties of such peptides can be costly (Oeller, Kang, Bolt, et al., 2023). To address this, CamSol-PTM was developed as a method that rapidly and reliably predicts the intrinsic solubility of peptides with mAA (Oeller, Kang, Bolt, et al., 2023).

2.8.2 Experimental solubility testing

Experimental solubility tests are typically conducted by measuring aqueous solubility with aliphatic alcohols such as methanol, ethanol, 1-propanol, and 2-propanol (Do et al., 2021). However, semipolar solvents such as dimethyl sulfoxide (DMSO) and dimethylformamide (DMF) or polar solvents such as water can also be used for solubility tests (Biswas et al., 2023).

Before testing the solubility of peptides in different solvents or aqueous solutions, the number and types of ionic charges of the peptide can be calculated. Depending on whether the peptide is acidic or basic, the solvent that facilitates the solubilization of the peptide can be chosen.

2.8.3 pH-dependent solubility

The solubility of a peptide occurs when the soluble and insoluble phases are in equilibrium. This solubility is dependent on pH, and pH, in turn, affects the stability of the peptide. Therefore, understanding the solubility of peptides at different pH levels is crucial. Computational methods such as CamSol can be employed for predicting the solubility variability of peptides and proteins across different pH conditions. This prediction is based on calculating partial charges using the Henderson-Hasselbalch equation, assessing hydrophobicity, and determining the residue pK (Oeller, Kang, Bell, et al., 2023).

The optimal pH for a peptide is one where both solubility and chemical stability are high. Buffers with high salt concentrations are used to control and maintain this pH (Bak et al., 2015).

2.9 Thermal stability

The evaluation of the thermal stability of a peptide is primarily conducted through thermogravimetric analysis (TGA) or differential scanning calorimetry (DSC). This assessment is important for understanding the energy required to destabilize the structures maintaining the peptide's conformation and how temperature changes may affect its biological activity. Moreover, for antimicrobial peptides (AMPs) intended for therapeutic purposes, knowledge of thermal stability is indispensable for determining synthesis and purification protocols, storage conditions, transportation requirements, and shelf life.

2.9.1 Differential scanning calorimetry (DSC)

Differential scanning calorimetry (DSC) is a technique that was developed in the 1960s (Chiu & Prenner, 2011). Primarily used to determine the energy of phase transitions, DSC measures the specific heat capacity of each thermally induced event (Chiu & Prenner, 2011). In the case of peptides and proteins, DSC is employed to measure the temperature of thermal transition and the energy required to destabilize the interactions that maintain the tertiary structure. This provides valuable insights into the thermodynamic properties and stability of the peptides (Durowoju et al., 2017).

One of the advantages of DSC is its sensitivity, and there are different models, such as Nano DSC, which facilitate studies at very high scanning rates with minimal sample sizes. It can operate at very low or very high temperatures (Chiu & Prenner, 2011).

DSC requires a reference sample, as it measures the difference in specific heat between the sample and the reference sample as a function of temperature (Jobin & Alves, 2019). This technique is commonly employed to analyze thermal denaturation, folding, and thermodynamic stability of proteins and peptides (Chiu & Prenner, 2011). The thermodynamic stability of peptides is associated with the equilibrium between native folded and unfolded states (Chiu & Prenner, 2011).

The thermodynamic properties of peptides and proteins obtained from DSC can be established by observing the different amounts of energy required to raise the temperature in the sample compared to the reference (Durowoju et al.,

2017). The energy needed to compensate for the temperature difference is then related to the thermodynamic properties of the biomolecule (Durowoju et al., 2017). Sample preparation for peptides involves a degassing process, and the reference sample can be the buffer used for peptide elution or the sample can be dialyzed against the buffer intended for use as a reference (Durowoju et al., 2017).

2.9.2 Thermogravimetric analysis (TGA)

The thermal characterization of peptides is commonly conducted using differential scanning calorimetry (DSC) or thermogravimetric analysis (TGA) (Dandurand et al., 2014; Golonka et al., 2023). The thermogravimetric analysis involves measuring changes in mass as a function of temperature, allowing the assessment of thermal stability, decomposition kinetics, and composition (Emiola-Sadiq et al., 2021). Comparative analyses of thermal stability and decomposition can be conducted among peptides, enabling a better understanding of their respective behaviors under varying temperature conditions (Adhikari et al., 2019).

Thermogravimetric analyses are commonly performed under a nitrogen atmosphere, utilizing lyophilized samples (Dandurand et al., 2014; Giri & Mandal, 2018).

The measurement of mass changes in thermogravimetric analyses is primarily carried out using two methods. First, the nonisothermal method involves heating the sample over a usually broad temperature range at a specific rate. Second, the isothermal method subjects the sample to a constant temperature for a specified duration (Moseson et al., 2020). The nonisothermal method is the most employed for analyzing the thermal stability and decomposition of peptides.

One of the primary advantages of TGA in the context of peptides is its rapid analytical capability and the minimal sample quantity required. Nevertheless, it serves as an approximate indicator of degradation, as the observed loss of mass may exceed the actual decomposition (Moseson et al., 2020).

2.10 Aggregation and oligomerization

Oligomerization and aggregation are critical considerations when proposing the use of peptides for therapeutic purposes, as they significantly influence biological response and effectiveness.

2.10.1 Thioflavin T (ThT) assay

Thioflavin T (ThT) is a dye composed of a dimethylaminobenzene derivative of benzothiazole, frequently utilized for labeling protein or peptide aggregates, especially those associated with amyloid peptides or proteins (Arad et al., 2020; Mao et al., 2011). The assay involves placing peptide-containing solutions in microplates with a 10 μ M ThT concentration, followed by fluorescence level measurements using a plate reader (Arad et al., 2020).

However, one of the main limitations is the lack of a quantitative relationship between fluorescence and aggregates, particularly fibrils, even though the assay is widely used for fibril aggregation analysis (Bourhim et al., 2007; Xue et al., 2017).

2.10.2 Transmission electron microscopy (TEM)

Transmission electron microscopy (TEM) is a technique that has been in use since the 1940s (Winey et al., 2014). It is a widely employed tool for the analysis of cellular components, including peptides and proteins (Winey et al., 2014). In the process of TEM analysis, samples are typically chemically fixed, dehydrated, and stained (Winey et al., 2014). The principle of this technique involves the use of a high-energy electron beam transmitted through the solid sample, resulting in images with higher resolution and magnification compared to optical microscopy (Stachowiak et al., 2004). One of the main advantages of TEM is the ability to elucidate the assembly, morphological features, and structural properties of aggregates (Goldsbury et al., 2011; Sung et al., 2015).

This technique allows for the characterization of protein or peptide aggregates using fluorescent markers (Anderson & Webb, 2011). It is crucial to ensure that the markers do not interfere with the aggregation reactions, as peptides and proteins can associate with various oligomeric structures (Anderson & Webb, 2011).

The study of peptide and protein aggregates is crucial, especially when they are employed for therapeutic purposes, as aggregates often tend to elicit unwanted immune responses (Kumar et al., 2020).

2.10.3 Atomic force microscopy (AFM)

Atomic force microscopy (AFM) involves the movement of a sharp tip attached to a cantilever in response to tip-surface interactions (Sinha Ray, 2013). This movement is measured by focusing a laser beam on a photodiode (Ray, 2013). This technique is highly powerful as it allows for the acquisition of 3D images of surfaces with molecular-level resolution (Dufrêne, 2002). AFM measurements are commonly performed in an aqueous medium for peptide and protein aggregates (Nirmalraj et al., 2020).

It is a highly useful technique for the analysis of peptide and protein aggregates, allowing for the resolution of complex conformational states associated with protein aggregation (Ruggeri et al., 2019). Primarily employed for the analysis of fibrillar aggregates (Nirmalraj et al., 2020; Xie et al., 2021), it serves as a precise tool for studying aggregates owing to its sub-nanometric instrumental resolution at the single-molecule level (Ruggeri et al., 2019). The introduction of this technology dates back to 1986 by Binnig, Quate, and Gerber (Ruggeri et al., 2019). The technique can be utilized in either dynamic or static modes, with a preference for dynamic mode due to the potential sample damage and introduction of artifacts associated with static mode frictional forces (Ruggeri et al., 2019).

2.10.4 Dynamic light scattering (DLS)

Dynamic light scattering (DLS) is a commonly used technique for measuring the size of aggregates (Pignataro et al., 2020). DLS involves measuring the fluctuation of scattered light (Pignataro et al., 2020). For peptide and protein aggregates, a continuous distribution fit of the autocorrelation function is typically performed due to the broad size distribution (Pignataro et al., 2020).

Samples to be analyzed by DLS are placed in quartz cuvettes, and it is recommended to perform multiple measurements to obtain accurate results (Shardlow et al., 2021). Additionally, it is crucial to know the refractive index and absorption value of the sample for DLS measurements (Shardlow et al., 2021). The hydrodynamic diameter of aggregates can be calculated using various mathematical equations such as the Stokes-Einstein equation and from intensity distributions (Shardlow et al., 2021). Generally, the polydispersity index is also measured, and a higher value indicates less precise measurements (Shardlow et al., 2021).

2.10.5 Fluorescence resonance energy transfer (FRET)

Fluorescence resonance energy transfer (FRET) is a widely employed technique for scrutinizing protein-protein or peptide-peptide interactions, rendering it applicable for observing protein or peptide aggregation processes (Wan et al., 2022). This method involves utilizing a fluorescent protein or peptide as a donor and another protein or peptide as an acceptor (Wan et al., 2022). In essence, FRET proves to be a valuable and complementary approach to atomic force microscopy, enabling a more in-depth analysis of intermolecular aggregation and assembly processes (Yang et al., 2018). In general, it is a highly sensitive technique that also enables the detection of oligomers. For instance, in a study, it was employed to detect oligomers formed by β -amyloid peptides (Wennmalm et al., 2015).

3 Biochemical characterization of AMPs

3.1 Pharmacokinetics

The term “pharmacokinetics” refers to the time course of a drug’s concentration in a bodily fluid—ideally plasma or blood—following the delivery of a specific dosing regimen. It includes all procedures pertaining to medication distribution, metabolism, excretion, and absorption. Any drug, including peptides and proteins, must comprehend the dose-concentration-effect relationship since it establishes the basis for clinical application and dosing regimen design (Meibohm, 2019).

Yet, peptides’ pharmacokinetics pose extra hurdles due to certain distinctive characteristics: (i) their definition through the production process within a living organism, as opposed to a chemically precise structure and purity typical of small-molecule drugs; (ii) their structural resemblance to endogenous proteins and nutrients; (iii) their close involvement in molecular-level physiological processes, often incorporating regulatory feedback mechanisms; and (iv) the analytical difficulties in detecting and measuring them amidst numerous similar molecules (Meibohm, 2019).

The analysis of short peptides can be conducted using either immunological assays or LC-MS/MS methods. Immunological assays, although utilized for their bioanalysis, exhibit limited selectivity due to the lack of antibody specificity (Ewles & Goodwin, 2011; Hoofnagle & Wener, 2009; Pan et al., 2009; van den Broek et al., 2013). Additionally, the

production of specific antibodies for novel peptides can be time-consuming (Ewles & Goodwin, 2011). Conversely, liquid chromatography coupled with tandem mass spectrometry (LC-MS/MS) offers both chromatographic separation of analytes and selective mass-based detection. Particularly, LC-MS/MS analysis of short peptides requires minimal multiple charging to achieve selective ionization of a specific m/z value. However, LC-MS/MS methods present various inherent challenges in peptide bioanalysis, including nonspecific adsorption, matrix suppression, and stability issues (Becher et al., 2006; Ewles & Goodwin, 2011; Ji et al., 2003; Tamvakopoulos, 2007; van den Broek et al., 2013). Despite these challenges, some researchers argue that LC-MS/MS could be the preferred approach for sensitive, selective, and high-throughput analysis (Ewles & Goodwin, 2011; Puttreu et al., 2019).

3.1.1 *In vivo Studies*

The pharmacokinetics of AMPs has garnered significant attention in recent research endeavors. Various studies have delved into understanding the *in vivo* dynamics of these peptides (Ali et al., 2009; Cui et al., 2021; Holfeld et al., 2015; Huang et al., 2022; Puttreu et al., 2019; Stone et al., 2019; Xiong et al., 2011). Cui and collaborators, for instance, developed a comprehensive pharmacokinetic study of the peptide OM19r-8 its modified pegylated variant, mPEG5-OM19r-8, utilizing reverse-phase high-performance liquid chromatography (RT-HPLC) for peptide quantification in plasma. Noncompartmental methods were applied to analyze the pharmacokinetic parameters, elimination of half-life, area under the drug concentration-time curve (AUC), mean residence time, clearance, and apparent volume of distribution (Cui et al., 2021).

Similarly, Ali and collaborators (Ali et al., 2009) explored the pharmacokinetics of novel antimicrobial cationic peptides, NAB 7061 and NAB 739. The concentration of NAB 7061 in plasma and urine was determined by HPLC using a modified method for colistin (Li et al., 2003). A liquid chromatography (LC)/electrospray ionization (ESI) mass spectrometry (MS) method was developed and validated to quantify NAB 739 in plasma and urine samples. Noncompartmental analyses of NAB 7061 and NAB 739 in plasma were conducted for their pharmacokinetics. The pharmacokinetic parameters evaluated were body clearance, volume of distribution at steady state, and terminal half-life.

Puttreu and collaborators (Puttreu et al., 2019) developed a selective liquid chromatography-tandem mass spectrometry (LC-MS/MS)-based bioanalytical method to uncover the pharmacokinetic aspects of the novel cationic antimicrobial peptide S016-1271. They determined parameters such as the area under the curve, half-life, clearance in the central compartment, clearance in the peripheral compartment, volume of distribution in the central compartment, volume of distribution in the peripheral compartment, mean residence time, and volume of distribution at steady state.

Additionally, Xiong and collaborators (Xiong et al., 2011) evaluated the novel plectasin-derived cationic antimicrobial peptide antibiotic NZ2114. Peptide plasma concentrations were determined by using a qualified liquid chromatography/tandem mass spectrometry (LC/MS-MS) assay. Apparent half-life, $C_{\max}$, and AUC pharmacokinetic parameters were presented.

Furthermore, Holfeld and collaborators (Holfeld et al., 2015) evaluated proline-rich antimicrobial peptides (PrAMPs) Onc72 and Api88 *in vivo* metabolic stability and distribution for the first time. Monoclonal IgG1/IgG2b antibodies produced by hybridoma cells were mapped to different Onc72 regions and combined in a sandwich-ELISA in a pharmacokinetic study (there was no production of monoclonal antibodies (mAbs) against Api88). The identified epitopes of anti-Onc72 mAbs allowed the quantification of Onc72 in murine plasma, organ-homogenates (liver, kidney, and brain), and urine. These studies collectively contribute to advancing our understanding of AMP pharmacokinetics and underscore the diverse methodologies employed.

3.1.2 *Data analysis*

Pharmacokinetic data analysis offers two main methodologies: nonlinear regression analysis for model fitting and noncompartmental analysis (NCA). When the primary objective is to quantify exposure levels postpeptide administration (e.g., AUC) and to assess associated pharmacokinetic parameters such as clearance, elimination half-life, $T_{\max}$, and $C_{\max}$, NCA typically emerges as the preferred methodology. This preference stems from NCA's advantage of relying on fewer assumptions compared to model-based methods (Gabrielsson & Weiner, 2012).

Generally, NCA directly determines key parameters as follows:

- $C_{\max}$: Maximum observed concentration (concentration)
- $T_{\max}$: The time when the maximum concentration was observed (time)
- AUC: The area under the curve (time $\times$ concentration)
- AUMC: The area under the first moment curve (time² $\times$ concentration)

Subsequently, essential properties such as clearance, steady-state volume of distribution, and terminal half-life can be calculated using derived equations:

- **Clearance:** $CL = \frac{\text{Dose}}{AUC}$
- **Mean residence time:** $MRT = \frac{AUMC}{AUC}$
- **Steady-state/apparent volume of distribution:** $V_{ss} = MRT \times CL$
- **Elimination of half-life:** Terminal slope of the natural log of the data

Commonly used software for NCA includes Drug and Statistics (DAS) software, Phoenix WinNonlin professional software, and the PKNCA package, an R package specifically designed for NCA calculations. Additionally, various other software options are available for pharmacokinetic data analysis, offering flexibility and versatility in approach (<https://www.pharmpk.com/soft.html>).

3.2 Bioavailability and stability

Coupled with pharmacokinetic investigations, assessments of bioavailability and stability are essential components. Some stability analyses, such as plasma stability, can essentially be viewed as in vitro pharmacokinetic studies (Puttreu et al., 2019). To improve peptide stability and in vivo bioavailability, a variety of chemical modifications are employed to mitigate AMP degradation. These modifications commonly include N-acetylation, C-amidation, cyclization, substitution of unnatural amino acids or analogs, prodrug design, PEGylation, lipopeptide synthesis, and peptidomimetics (Shao et al., 2019). Confirming that peptide bioactivity remains unaltered often involves monitoring plasma levels or evaluating antimicrobial efficacy. Understanding peptides' bioavailability, absorption, susceptibility to enzymatic degradation, and storage stability is crucial, as it sheds light on their behavior in practical applications. The therapeutic potential of peptides heavily relies on their ability to remain intact until reaching the target organ. This emphasizes the importance of investigating bioactive peptides as prospective therapeutic agents, despite their limited utility due to high instability, rapid clearance from the bloodstream, low membrane permeability, and vulnerability to gastrointestinal digestion (Arhewoh et al., 2011).

3.2.1 Peptide bioavailability improvement

Numerous strategies have been developed to enhance peptide activity, prolong their half-life, and improve their distribution, thereby enhancing tissue permeability. Prodrug peptides, which are pharmacologically inert, temporarily alter a drug's physicochemical properties to enhance efficacy or reduce toxicity. Peptidomimetics, incorporating nonpeptide structural elements, mimic or counteract the biological functions of natural peptides. Structural modifications, including pegylation with polyethylene glycol (PEG) molecules, lipidation with fatty acids, isoprenoids, or cholesterol, and glycosylation with carbohydrates, are also utilized. Additionally, peptides undergo chemical alterations such as cyclization via disulfide bond formation between cysteine residues, amino acid substitutions, and N-methylation (Segura-Campos et al., 2011; Wang, Hong, et al., 2021; Zhu et al., 2021). These modifications attenuate enzymatic degradation, enabling peptides to circulate in the bloodstream intact, thereby maximizing their efficacy through prolonged availability, heightened potency, and/or enhanced catalytic activity, depending on the specific molecule (Biron et al., 2008).

The strategy of employing enzyme inhibitors and absorption enhancers to protect peptides from enzymatic degradation and overcome inherent challenges such as hydrophilicity, molecular size, and charge has become foundational in protein delivery. Additionally, targeting specific absorption sites and refining dosage forms, such as lipid vesicles or colloidal carrier systems with or without mucoadhesive polymers, present promising avenues for enhancing peptide absorption (Lee, 2002).

3.2.2 Plasma stability

Comprehending and enhancing the stability of peptides within physiological environments, notably plasma, holds crucial significance as it profoundly impacts the effectiveness of AMPs. Assessing the stability of peptides in plasma serves as a pivotal criterion in evaluating the potential therapeutic value of these compounds. The complex composition of plasma, featuring enzymes, proteases, and diverse biomolecules, poses considerable hurdles to the stability of AMPs, potentially curtailing their therapeutic effectiveness. Certain investigators have investigated chemical modifications as a means to improve peptide stability in the bloodstream (Wang, Zou, et al., 2021).

In addition to conducting pharmacokinetic investigations on the novel cationic antimicrobial S016-1271 peptide, Puttreu and collaborators (Puttreu et al., 2019) devised a plasma stability assay to evaluate the proteolytic resistance

of the peptide in plasma. This assay, performed using both mouse and human plasma, involved sample processing through solid-phase extraction (SPE), followed by analysis using liquid chromatography-tandem mass spectrometry (LC-MS/MS). The findings revealed that the peptide demonstrated remarkable stability.

Wang and collaborators (Wang, Zou, et al., 2021) conducted a series of modifications on the antimicrobial peptide F_{2,5,12}W, which included alterations in the amino acid sequence, introduction of cysteine and other common amino acids, development of peptide dimers via disulfide bonds, and conjugation with methoxy polyethylene glycol (mPEG). Their aim was to enhance antimicrobial activity and plasma stability and reduce hemolytic activity. While mPEG conjugation notably improved plasma stability and reduced hemolytic activity, it concurrently decreased antimicrobial activity. Interestingly, altering the peptide sequence without changing its amino acid composition had a minimal impact on both antimicrobial activity and plasma stability. The introduction of cysteine notably enhanced plasma stability but also increased peptide hydrophobicity, leading to elevated hemolysis.

Also, Chen and collaborators (Chen et al., 2021) developed a straightforward de novo designed peptide, pepD2, comprising only three types of amino acid residues (W, K, L). They utilized reversed-phase high-performance liquid chromatography (HPLC) to evaluate peptide integrity in plasma over time. Remarkably, the D-form peptide, pepdD2, where the D-enantiomers replaced the L-enantiomers of Lys(K) and Leu(L) residues, demonstrated more than a 12-fold extension in peptide half-life in plasma, highlighting enhanced stability properties.

3.2.3 Enzymatic degradation, pH, temperature, serum and salt stability

Various factors influence the effectiveness of AMPs, encompassing enzymatic degradation, pH variations, temperature changes, exposure to serum, and salt concentrations, all of which are crucial in determining their efficacy. Within biological systems, AMPs are prone to degradation by a range of proteolytic enzymes, swiftly breaking down these molecules into inactive fragments. To mimic in vivo conditions, enzymatic profiling assays are performed to gauge the susceptibility of AMPs to degradation by key proteases such as trypsin, chymotrypsin, and elastase.

The pH of the surrounding environment profoundly impacts the structure and function of AMPs. With diverse pH conditions encountered at different anatomical sites, ranging from the acidic environment of the stomach to the neutral pH of blood and tissues, pH stability becomes a critical factor influencing AMP bioactivity. Temperature fluctuations also play a significant role in evaluating AMP stability. Within the body, AMPs encounter a wide range of temperatures. Studies on temperature stability provide crucial insights into the thermal resilience of AMPs, aiding in the assessment of their stability and activity under physiologically relevant conditions. Moreover, the presence of serum and salt concentrations in biological fluids can impact the stability and efficacy of AMPs. Serum proteins may interact with AMPs, potentially altering their pharmacokinetic properties or susceptibility to proteolytic degradation. Similarly, elevated salt concentrations can influence the conformation and membrane interactions of AMPs, thereby affecting their antimicrobial activity (Aoki & Ueda, 2013; Arias et al., 2018; Mohammad et al., 2015).

In their pharmacokinetic investigation of the N-terminal PEGylated D-form of the antimicrobial peptide OM19r-8, developed by Cui and colleagues (Cui et al., 2021), the researchers conducted a comprehensive assessment of the peptide's enzymatic, pH, temperature, serum, and salt stability. Additionally, they determined the Minimum Inhibitory Concentrations (MICs) of the polypeptide against *Escherichia coli* ATCC25922. The findings revealed a notable enhancement in the resistance of the antibacterial peptide OM19r-8 to protease degradation, with a surge of 4–32-fold in MIC values. Furthermore, the modified peptide exhibited improved stability under varying conditions, including temperature, trypsin exposure, caroid presence, serum (at 50% concentration), and salt.

Meanwhile, Kim and collaborators (Kim et al., 2014) devised a series of peptides comprised of 15 unaltered natural amino acids while meticulously preserving structural features crucial for the antimicrobial activity and selectivity of AMPs. These peptides were strategically arranged to minimize their susceptibility to protease-mediated cleavage. Subsequent incubation of each peptide with trypsin, chymotrypsin, or aureolysin was conducted to assess their susceptibility to proteolytic digestion. Evaluation via SDS-PAGE analysis and growth inhibition assay unveiled that while the antimicrobial activity remained unaffected by the enzymes, cleavage of peptides was observed, emphasizing the importance of protease resistance in peptide design for sustained antimicrobial efficacy.

3.2.4 Storage stability

The biochemical assessment of AMPs encompasses a comprehensive evaluation of their resilience across diverse storage environments, encompassing factors such as temperature, humidity, pH, and light exposure. These analyses are indispensable for ascertaining the practical applicability and durability of AMPs in pharmaceutical or industrial settings. Storage stability investigations typically entail the continuous monitoring of multiple parameters over time, including peptide

concentration, secondary structure, antimicrobial efficacy, and the presence of degradation byproducts. Various influences can impact peptide stability during storage, ranging from fluctuations in temperature, humidity, and pH to exposure to light, notably UV radiation.

In the study conducted by Puttreu and collaborators (Puttreu et al., 2019), the stability of S016-1271 in plasma was systematically investigated under diverse conditions. The assessment encompassed bench-to-stability evaluations (ambient temperature for 6 h), freeze-thaw stability tests (-80°C , three cycles of freeze-thaw), long-term stability investigations (-80°C for 30 days), and autosampler stability examinations (6°C for 24 h). Results indicated that the accuracy and precision of stock solution stability samples remained within accepted parameters. Furthermore, the percentage deviation from the nominal concentration under varied storage conditions in both mice and plasma samples fell within acceptable ranges, suggesting the stability of the analytes under various storage conditions.

3.3 Peptide-lipid interactions

As described by Lāce and collaborators (Lāce et al., 2023), the interactions between AMPs and membranes involve disruption (Sato & Feix, 2006), translocation (Ulmschneider, 2017), and permeabilization (Hollmann et al., 2018). Initially, AMPs adsorb onto the outer membrane surface, causing an imbalance between the inner and outer lipid layers. This leads to a drive for an equal distribution of peptides between the layers. Hydrophobic peptides may incorporate into the lipid bilayer, translocating across without causing leakage. Many AMPs disrupt prokaryotic membranes by forming pores in the lipid bilayer. This can occur via different mechanisms, such as barrel-stave, toroidal, or carpet/detergent. Regardless of the pathway, membrane disruption results in cell interior leakage and eventual cell death (Sani & Separovic, 2016; Sengupta et al., 2008). Peptides not disrupting membranes can still be considered AMPs if they induce cell damage intracellularly, affecting both prokaryotic and eukaryotic cells (Benfield & Henriques, 2020; Cardoso et al., 2019; Takeshima et al., 2003).

Galdiero and collaborators have extensively outlined a range of techniques available for investigating the morphological changes induced by peptides on membranes, as well as for illustrating interactions and describing both peptide and lipid structure and dynamics (Galdiero et al., 2013). Morphological changes can be evidenced by electron microscopy (EM) and atomic force microscopy (AFM) offering high spatial resolution, but sample preparation can introduce artifacts. Classical fluorescence imaging, while sensitive, may perturb the system and has limited resolution (El Kirat et al., 2010; García-Sáez et al., 2007; Oreopoulos et al., 2010; Volinsky et al., 2006). Additionally, calorimetry can identify thermodynamic changes in the membrane (Lohner & Prenner, 1999), while solid-state nuclear magnetic resonance (NMR) and X-rays provide insights into global phase changes (Faucon et al., 1995; Lohner & Prenner, 1999; Pott et al., 2001).

Fluorescence, electron paramagnetic resonance spectroscopy (EPR), IR spectroscopy, circular dichroism (CD), surface plasmon resonance (SPR), X-rays, and NMR techniques are capable of illustrating interactions and describing both peptide and lipid structure and dynamics. IR, CD, and EPR studies can reveal secondary peptide structure, degree of penetration, and orientation (Castano & Desbat, 2005; Jean-François et al., 2007). NMR uniquely provides both topology and three-dimensional structure information on peptides in membranes (Bechinger et al., 2011; Resende et al., 2009). Computational methods are also employed for studying peptide-membrane interactions, offering insights into dynamic processes not accessible through experiments alone. However, these simulations have limitations, such as overlooking peptide-peptide interactions and gradients of mobility and dielectric constant across the bilayer. Despite the array of experimental techniques available, obtaining comprehensive information on peptide-membrane interactions remains challenging. Interpretation of data often requires a model, which can itself be challenging to construct. Therefore, a combination of complementary experimental techniques is necessary to derive a complete understanding of these interactions (Galdiero et al., 2013).

3.3.1 Membrane permeation

Several *in vitro* models, such as artificial lipid bilayers, cell monolayers, or membrane vesicles, are typically employed to evaluate how peptides interact with biological and artificial barriers, aiding in the prediction of their properties (Deshayes et al., 2022; Kim et al., 2021; Ouyang et al., 2022; Zakany et al., 2023). For instance, the membrane leakage fluorescence assay, which relies on vesicles loaded with self-quenching dyes, serves as a widely adopted method to quantify the activity of AMPs capable of permeabilizing membranes and inducing leakage of encapsulated contents. This technique leverages the self-quenching nature of the dye, which is initially entrapped at high concentrations. The subsequent release of the dye from the vesicles into the surrounding media results in its dilution, thereby causing a measurable increase in fluorescence intensity (Hollmann et al., 2018).

In a study by Flaten and collaborators (Flaten et al., 2011), the permeation of cationic antimicrobial di- and tripeptides derived from lactoferricin via interaction with the human intestinal peptide transporter hPEPT1 and via passive routes. The ability of the peptides to passively permeate across membranes was investigated using the phospholipid vesicle-based permeation assay (PVPA). The passive permeation of these peptides across membranes was investigated using the phospholipid vesicle-based permeation assay (PVPA), where UV absorbance at appropriate wavelengths for the peptides was employed to quantify their presence in the acceptor compartment. Furthermore, Moniruzzaman, Zahidul Islam and collaborators (Moniruzzaman et al., 2017) delved into the interactions of Lactoferricin B (Brunetti et al., 2016; Lee et al., 2016; Mansour et al., 2014; Mygind et al., 2005; Wakabayashi et al., 1996; Yamauchi et al., 1993) with *E. coli* cells and giant unilamellar vesicles (GUVs). *E. coli* containing calcein and GUVs containing AlexaFluor 647 hydrazide were employed, and leakage was determined via confocal microscopy.

3.3.2 Surface morphology

Understanding the surface morphology of eukaryotic cells, including bacteria and yeast, is crucial for unraveling the mechanisms of action behind AMPs. Eukaryotic cells are enveloped by membranes composed of phospholipid bilayers, which are essential for their structural integrity and function. Morphological analysis offers insights into how these membranes are affected by AMP treatment, revealing potential disruptions caused by the peptides' potent permeabilizing activity (Lāce et al., 2023). The outermost layers of microbial cells play a pivotal role as the primary interface for AMP interactions, influencing the peptides' efficacy in disrupting cellular integrity and exerting antimicrobial effects (Kim et al., 2014).

Prince and collaborators (Prince et al., 2016) evaluated the lipid-II independent antimicrobial mechanism of the AMP Nisin depending on its crowding and degree of oligomerization. They determined bacterial (*B. subtilis* and *E. coli*) surface morphology changes via field emission scanning electron microscopy (FE-SEM). They employed low and high concentrations of the peptide to compare the surface morphological changes. Electron microscopy images showed that high peptide concentration resulted in a fivefold reduction in the surface area of *B. subtilis* and significant bleb-like membrane protrusions in *E. coli*.

Furthermore, Torcato and collaborators (Torcato et al., 2013) determined the mechanism of action of the AMP BP100 analogues by direct bacterial imaging with atomic force microscopy (AFM). They determined the bacterial cell surface roughness (R_{rms}) and Changes in bacterial shape (S_{rms}). Membrane disruption was observed on *S. aureus* for all the peptides at the highest concentration tested.

3.3.3 Lipopolysaccharides (LPS) interaction

Investigating the affinity of AMPs to LPS is a widely used assay to characterize peptide's mechanism of action (Bruschi et al., 2010; Torcato et al., 2013). LPS, found in the outer membrane of Gram-negative bacteria, plays a pivotal role in bacterial pathogenesis and is a prime target for AMPs due to its essential role in maintaining bacterial integrity and virulence. By probing the interactions between AMPs and LPS, researchers can unravel the molecular mechanisms underlying the antimicrobial activity of these peptides, including membrane disruption and pore formation.

Bruschi and collaborators (Bruschi et al., 2010) evaluated the insertion of SB041 into monolayers formed of LPS from *E. coli* or *Pseudomonas aeruginosa* as an indication of the peptide's ability to bind endotoxin and penetrate the Gram-negative's outer membrane. They followed the increment in the surface pressure of the LPS film upon intercalation of the peptide. An additional time-kill assay was assessed to evaluate whether endotoxin could influence the peptide's antimicrobial activity. Measurement of the SB041 ability to bind LPS was also performed by a fluorescent displacement assay. The results indicated that SB041 exhibited strong binding to purified LPS and demonstrated facile penetration into LPS monolayers. However, the authors noted that while SB041 showed binding affinity to LPS, this did not necessarily correlate with LPS-neutralizing activity, suggesting that the interactions between SB041 and LPS may be of a complex nature.

3.3.4 Molecular dynamics simulation

Molecular dynamics (MD) simulations play a crucial role in understanding membrane destabilization caused by peptide action. This computational approach offers an efficient means to investigate the properties of biological membranes (La Rocca et al., 1999; Sengupta et al., 2008). By simulating the interaction between peptides and lipid bilayers, MD simulations can provide valuable insights into how these peptides influence membrane activity, and conversely, how the lipid bilayer affects peptide behavior (Hasper et al., 2004). The results obtained from MD simulations, such as area per lipid, root mean square deviation (RMSD), and deuterium order parameter, can be compared with experimental observations from techniques such as X-ray and NMR to further understand the impact of peptide interactions on lipid acyl chains within model membranes (Nagle & Tristram-Nagle, 2000; Prince et al., 2016). These comparisons help validate the computational models and provide a comprehensive understanding of membrane dynamics.

Cromacs is the most employed software that uses command line interfaces for MD (Aghazadeh et al., 2020; Appelt et al., 2005; Khatami et al., 2014; Lai & Kaznessis, 2018; Prince et al., 2016; Wang et al., 2014). Examples of software that adopt the graphical user interfaces type are SwissPDB viewer (Dutta et al., 2017; Naha & Ramaiah, 2023; Porto et al., 2020), Pymol (Antunes et al., 2013; Azhagiya Singam et al., 2014; Pimthon et al., 2009), VMD (Allsopp et al., 2022; Basu et al., 2022; Duay et al., 2019; Petkov et al., 2019; Talandashti et al., 2021), USCF Chimera (Guntamadugu et al., 2023; Ranade & Ramalingam, 2020), among others. An example of modeling web servers employed for MD is Swissmodel (Mehta et al., 2022).

Prince and collaborators (Prince et al., 2016) also developed an analysis of membrane destabilization involving MD simulations to investigate the properties of biological membranes. This approach proved efficient in providing insights into the interaction between peptides and lipid bilayers, as well as the reciprocal influence on membrane activity. Simulations of lipid membranes, including pure POPC and mixed bilayers (PC/PE/PS), were conducted using the Gromacs package with the Gromos 53a6 (Berendsen et al., 1995) force field. Parameters such as deuterium order parameter, density of phosphate head-groups and acyl chains, RMSDs of atomic positions, and water defects were computed throughout the simulations. The analysis included plotting the order parameter versus carbon atom position, density calculations along the Z-dimension of the simulation box, and determination of area per lipid and membrane curvature using specific tools.

3.4 Receptor affinity techniques

The identification of possible therapeutic targets is facilitated by receptor binding studies, which provide insightful information about the precise interactions between peptides and cellular targets. It is important to evaluate how AMPs interact with cell membranes, and binding affinity assays can help with this. The lipid binding affinity to the membrane can be clarified using methods such as surface plasmon resonance, fluorescence polarization, and radioligand binding assays. This information is crucial for comprehending the interactions between AMPs and membranes.

Affinity binding assays are commonly employed to assess the strength of interaction between a ligand and its target biomolecule, typically represented by the equilibrium dissociation constant. Surface plasmon resonance spectroscopy is a label-free technique rapidly gaining prominence for studying ligand binding interactions with membrane proteins, offering real-time quantitative measurements of binding affinities and kinetics. This technique requires immobilizing one binding component on a sensor chip while the other component in the solution is flowed over the sensor surface, with binding interactions detected through changes in refractive index (Douzi, 2017). Fluorescence polarization technology, on the other hand, relies on measuring molecule rotation and has found widespread use in studying molecular interactions in solution, particularly suitable for measuring binding and dissociation between molecules when one of the binding partners is relatively small and fluorescent (Du, 2015).

3.4.1 Radioligand binding assay

Radioactive ligands are frequently employed to assess ligand binding to receptors, with assays measuring the interaction of a radiolabeled ligand, such as a peptide, with cells or cell membranes housing the target receptor. Such assays enable the construction of saturation curves, competition studies, and kinetic experiments. Criteria for selecting a suitable radioligand include high specific activity, low nonspecific binding, high purity, and selectivity (Bosma et al., 2019; Maguire et al., 2012). In radioligand binding assays, detection of binding is accomplished using techniques such as scintillation, which detects light emission subsequent to radiation absorption, or autoradiography. This methodology yields highly sensitive measures of binding affinity.

Chan and collaborators (Chan & Gallo, 1998) devised a radioligand binding assay to investigate the association of PR-39, a proline-arginine-rich antimicrobial peptide, with cells. PR-39 was iodinated via the IODO-GENTM bead technique (I-PR-39). The gamma counter was used to quantify I-PR-39 binding to cell proteins, with specific binding determined by the discrepancy in counts between cells with and without excess unlabeled PR-39. Their findings indicated that PR-39 binds to NIH 3 T3 cells in a saturable manner, indicative of a binding target's existence. Similar to full-length PR-39, PR-39 (Sarfo et al., 2003) also interacts with lipid bilayers. Upon interaction with the membrane, PR-39 (Sarfo et al., 2003) swiftly enters human microvascular endothelial cells and binds various cytoplasmic proteins. PR-39 selectively binds recombinant SH3-containing proteins and was observed to bind a native SH3-containing protein, p130^{Cas}. Treatment with PR-39 (Sarfo et al., 2003) alters p130 localization in endothelial cells. These results underscore that PR-39 (Sarfo et al., 2003) binds to an SH3-containing signal transduction molecule, potentially elucidating the diverse effects PR-39 exerts on mammalian cell behaviors.

3.4.2 Surface plasmon resonance

Surface plasmon resonance (SPR) leverages the optical phenomenon of SPR to monitor real-time changes in mass occurring on a sensor chip surface. Innovative surface chemistry techniques have been devised to immobilize lipid bilayers, enabling the study of protein-membrane interactions in real time (Hall & Aguilar, 2010). SPR spectroscopy exploits the surface plasmon resonance phenomenon, facilitating label-free monitoring of biomolecule binding to biomimetic surfaces by detecting alterations in adsorbed mass at the sensor surface (Hall et al., 2003). By measuring changes in refractive index resulting from molecular binding to a metal-coated surface, surface plasmon resonance (SPR) can assess binding kinetics, including association and dissociation rates, as well as binding affinity. This technique has found extensive application in investigating the membrane-binding behavior of AMPs (Bruschi et al., 2010; Kamimori et al., 2005; Kinouchi et al., 2014; Mozsolits & Aguilar, 2002).

Mozsolits and collaborators (Mozsolits et al., 2001) examined the lipid binding characteristics of antimicrobial peptides, including magainin 1, melittin, and the C-terminally truncated melittin analogue (21Q), utilizing SPR with a hybrid bilayer membrane system. They employed linearization analysis and curve fitting through numerical integration to derive estimates for association (k_a) and dissociation (k_d) rate constants. Their findings revealed distinct lipid preferences among the peptides, with magainin 1 showing a preference for negatively charged dimyristoyl-l- α -phosphatidyl-dl-glycerol (DMPG), melittin interacting with both zwitterionic dimyristoyl-l- α -phosphatidylcholine and anionic DMPG, and the 21Q analogue exhibiting lower binding affinity for both lipids. This highlighted the significant influence of the positively charged C-terminus of melittin on its membrane binding properties.

3.4.3 Fluorescence polarization

Fluorescence polarization (FP) is an effective method for investigating molecular interactions by observing alterations in the apparent size of molecules labeled with fluorescence or inherently fluorescent substances, commonly known as tracers or ligands (Checovich et al., 1995; Heyduk et al., 1996; Jameson & Sawyer, 1995; Nasir & Jolley, 1999). Unlike other techniques for assessing molecular binding, FP stands out for its ability to provide a direct and almost immediate assessment of the ratio between bound and free tracers.

FP allows equilibrium analysis down to low picomolar concentrations, with the capability to monitor changes in binding caused by factors such as pH, temperature, and salt concentration without affecting samples. It also facilitates kinetic analysis of association and dissociation reactions. Unlike methods requiring the separation of bound and free species, FP is homogeneous, saving time and maintaining reaction equilibrium for accurate quantification of binding. Alternative techniques such as fluorescence resonance energy transfer (FRET) and time-resolved fluorescence (TR-FRET) are more complex (Heyduk et al., 1996), requiring multiple labeling reactions compared to FP's single-label approach.

The terms “polarization” and “anisotropy” are often used interchangeably in FP literature, both derived from measured vertical and horizontal intensities. While they contain the same information, anisotropy values are often preferred due to easier mathematical manipulation. Despite its advantages, FP has limitations such as susceptibility to local motion effects, dye lifetime constraints, and restrictions based on ligand size and molecular weight. Additionally, it may suffer from auto-fluorescence interference (Hall et al., 2016).

Do and collaborators (Do et al., 2006) pioneered FP assays of the human melanocortin 4 receptor (MC4R) using TAMRA-NDP-MSH as a tracer. Their study demonstrated the interaction between MC4R and 11-residue antimicrobial peptides derived from the Korean frog, *Rana rugosa*, showcasing the utility of FP in elucidating such molecular interactions.

3.5 Antiinflammatory and immunomodulatory properties

Some AMPs, beyond their established antibacterial, antifungal, and antiviral properties, exhibit additional antiinflammatory and immunomodulatory capabilities, categorizing them as innate defense regulator (IDR) peptides. These remarkable molecules possess the ability to modulate the body's immune response by various means. For instance, IDR peptides can enhance leukocyte migration to the infection site, promote macrophage differentiation and activation, facilitate neutrophil chemotaxis, induce mast cell degranulation and activation, modulate the production of chemokines and cytokines, and even stimulate angiogenesis and wound healing processes. The multifaceted immunomodulatory functions of IDR peptides are currently under extensive investigation and development. Exploring and harnessing the potential of IDR peptides as an indirect tool against infectious diseases represents a novel strategy for combating antimicrobial resistance. By delving deeper into their mechanisms of action and therapeutic applications, IDR peptides could revolutionize the approach to addressing the challenges posed by resistant pathogens, offering new avenues for treatment and disease management (Chessa et al., 2020; Guryanova & Ovchinnikova, 2022; Haney & Hancock, 2013; Lesiuk et al., 2022; Pizzo et al., 2018).

For instance, the host defense peptide S100A7 exemplifies this dual functionality, showcasing not only potent antibacterial, antibiofilm, antifungal, and antiparasitic actions but also profound effects on immune cells. Experimental assays have revealed that S100A7 induces chemokine expression in both B and T cells, while in monocytes, it stimulates the production of chemokines such as CXCL8, CCL2, and CCL7, along with antiinflammatory cytokines, and facilitates the differentiation into proinflammatory macrophages. Remarkably, S100A7 also exerts a suppressive effect on proinflammatory macrophages, exemplifying its intricate immunomodulatory role (Hancock et al., 2016). Significantly, there are numerous host defense peptides with this kind of dual functionality, S100A7 being just one. Further highlighting the potential of AMPs in therapeutic applications are others, such as HNP1, HNP2, HNP3, HNP4, HD5, HD6, HBD1, HBD2, HBD3, and HBD4, which exhibit significant immunomodulatory effects in addition to their antimicrobial properties (Hancock et al., 2016). Novel approaches to treating immune-related disorders and infectious diseases may be discovered via further investigation into the immunomodulatory profiles of these peptides, providing encouraging opportunities for clinical research (Lesiuk et al., 2022).

Brunetti and colleagues conducted a comprehensive assessment of the immunomodulatory and antiinflammatory properties of the novel antimicrobial peptide SET-M33, as detailed in their study (Brunetti et al., 2016). Notably, SET-M33 exhibited robust efficacy against Gram-negative bacterial infections, positioning it as a promising candidate for the treatment of lung infections caused by such pathogens, a critical medical need. Their investigation revealed compelling evidence of SET-M33's potent antiinflammatory activity, particularly evidenced by the significant reduction in the expression of various cytokines, enzymes, and signal transduction factors implicated in inflammation provoked by LPS derived from *P. aeruginosa*, *Klebsiella pneumoniae*, and *E. coli*. Utilizing both in vitro models involving macrophages and bronchial cells, including cells obtained from a cystic fibrosis patient, the researchers observed remarkable downregulation, up to 100%, of multiple inflammatory mediators upon stimulation with LPS and subsequent treatment with SET-M33, as validated through diverse molecular techniques such as RT-PCR, Western blotting, and Luminex technology. Moreover, in vivo experimentation illustrated the capability of SET-M33 to neutralize LPS, evidenced by a sharp reduction in tumor necrosis factor-alpha (TNF- α) levels in bronchoalveolar lavage of SET-M33-treated mice challenged with LPS compared to untreated counterparts. Additionally, the study elucidated an intriguing facet of SET-M33's functionality, demonstrating its profound capacity to stimulate keratinocyte migration in wound healing assays in vitro, highlighting its potent immunomodulatory attributes characteristic of innate immunity molecules. These findings underscore the multifaceted therapeutic potential of SET-M33, not only as an antimicrobial agent but also as a promising immunomodulatory agent with implications for the management of infectious and inflammatory conditions, heralding a new frontier in therapeutic development.

4 Conclusions

This chapter offers an extensive exploration of the principles, technical requirements, and limitations inherent in physicochemical and biochemical characterization techniques employed in the study of antimicrobial peptides (AMPs). Physicochemical techniques encompass a diverse array of methodologies, including chromatographic, spectroscopic, thermal, and microscopic approaches, alongside computational methods aimed at deciphering the relevant characteristics of AMPs. These techniques play a pivotal role in elucidating the structural, functional, and dynamic properties of AMPs, providing crucial insights into their mechanisms of action and potential applications.

In addition to physicochemical techniques, biochemical assays are instrumental in understanding the behavior of AMPs within biological systems. These assays offer valuable insights into the kinetics, stability, bioavailability, and interactions of AMPs with various biological structures, shedding light on their efficacy and safety profiles. However, it is imperative to acknowledge the inherent limitations and challenges associated with each analytical method, including instrument complexity, preparation requirements, and potential interferences, which may impact the reliability and interpretation of results.

Furthermore, the selection of an appropriate analytical method should be carefully tailored to the specific research objectives and sample characteristics to ensure accuracy and reproducibility. Researchers must navigate through a myriad of techniques, considering factors such as sensitivity, specificity, and robustness, to derive meaningful conclusions from their investigations.

Overall, a comprehensive understanding of physicochemical and biochemical characterization techniques is indispensable for researchers aiming to unravel the complexities of AMPs' functions and interactions in diverse biological contexts. By leveraging these analytical tools effectively, researchers can advance knowledge, foster innovation, and address pressing challenges in fields spanning pharmaceuticals, biotechnology, environmental science, and beyond.

AI Disclosure

During the preparation of this work, the author(s) used ChatGPT and QuillBot to review grammar and improve writing. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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ANTIMICROBIAL PEPTIDES

A Roadmap for Accelerating Discovery and Development

Edited by

Luis H. Reyes, Department of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia

Juan C. Cruz, Department of Biomedical Engineering, Universidad de los Andes, Bogotá, Colombia

Gregory R. Wiedman, Department of Chemistry and Biochemistry Seton Hall University, South Orange, NJ, United States

Contains a comprehensive description of computational and experimental approaches for the discovery, manufacture, and clinical translation of antimicrobial peptides.

Antimicrobial Peptides: A Roadmap for Accelerating Discovery and Development covers the most important efforts of scientists and engineers worldwide to accelerate the process of discovery, production, and eventual market penetration of more potent antimicrobial peptides. These efforts have been fueled by emerging technologies such as artificial intelligence and data science, molecular and CFD simulations, easy-to-use process simulation packages, microfluidics, and 3D printing, among many others. Such technologies can now be implemented and scaled up quickly and at relatively low cost in low-budget production facilities, critical to moving to sustainable and marketable products worldwide.

Discovering novel antimicrobial peptides rationally and cost-effectively has emerged as one of the significant challenges of modern biotechnology. Thus far, this process has been tedious and costly, resulting in molecules with activities far below those needed to address the current challenge of microbial resistance to antibiotics that takes the lives of thousands of people around the world every year. Due to these issues, large biopharmaceutical companies have decided not to invest in discovering novel antimicrobials, which is worrisome in the long run due to the possibility of a new pandemic.

This book also delineates how multidisciplinary teams have assembled to address the challenges of manufacturing, biological testing, and clinical trials to finally reach complete translation. It is an ideal resource for scientists, researchers, engineers, and students in the fields of biotechnology, microbiology, biochemistry, and pharmaceuticals. Finally, anyone interested in the emerging technologies that are enabling the discovery and development of new antimicrobial peptides may find this book interesting and informative.

Key Features

- Covers computational tools (including emerging artificial intelligence algorithms) and microfluidic systems for discovering and high-throughput screening of AMPs
- Discusses the application of bioprocess engineering scale-up approaches for AMPs production and purification with the aid of process simulation tools and rapid prototyping
- Highlights user-centered design and formulation of products with AMPs
- Describes the whole pipeline for AMPs production



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